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EmoAtlas: An emotional network analyzer of texts that merges psychological lexicons, artificial intelligence, and network science

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Abstract

We introduce EmoAtlas, a computational library/framework extracting emotions and syntactic/semantic word associations from texts. EmoAtlas combines interpretable artificial intelligence (AI) for syntactic parsing in 18 languages and psychologically validated lexicons for detecting the eight emotions in Plutchik's theory. We show that EmoAtlas can match or surpass transformer-based natural language processing techniques, BERT or large language models like ChatGPT 3.5 or LLaMAntino, in detecting emotions from Italian and English online posts and news articles (e.g., achieving 85.6% accuracy in detecting anger in posts vs the 68.8% value of ChatGPT and 89.9% value for BERT). EmoAtlas presents important advantages in terms of speed and absence of fine-tuning, e.g., it runs 12x faster than BERT on the same data. Testing EmoAtlas' and easily trainable transformers' relevance in a psychometric task like reproducing human creativity ratings for 1071 short texts, we find that EmoAtlas and BERT obtain equivalent predictive power (fourfold cross-validation, $\rho \approx 0.495$, $p < 10^{-4}$). Combining BERT's semantic features with EmoAtlas' emotional/syntactic networks of words gets substantially better at estimating creativity rates of stories ($\rho = 0.628$, $p < 10^{-4}$). This indicates an interplay between the creativity of narratives and their semantic, emotional, and syntactic structure. Via interpretable emotional profiles and syntactic networks, EmoAtlas can also quantify how emotions are channeled through specific words in texts, e.g., how did customers frame their ideas and emotions towards "beds" in hotel reviews? We release EmoAtlas as a standalone "text as data" computational tool and discuss its impact in extracting interpretable and reproducible insights from texts.

Keywords Cognitive networks · Artificial psychometrics · Affective computing · Text analysis · Frame semantics · Emotional analysis

Introduction

Text is a form of structured data. Words, punctuation and symbols (e.g., emojis) can be combined in text to convey ideas and emotions in ways that can be detected by humans (Carley, 1994; Fillmore & Baker, 2001) and, more recently, by artificial intelligence (AI) (Manning et al., 2014; Srivastava et al., 2022; Colladon et al., 2020; Stella, 2022). Whereas decomposing text in these elements is relatively easy, understanding the connections between words, ideas, and emotions is quite daunting (Gentzkow et al., 2019). Each word in isolation can be considered as a cue eliciting one or multiple emotions (Mohammad & Turney, 2013). To express emotions more strongly, a given text can feature many words eliciting the same emotion (Mohammad, 2012). In turn, this can change the intensity with which a given text elicits one or more emotions (Strapparava & Mihalcea, 2007). The collection of emotions elicited with different intensities by a text represents the "emotional profile" of the latter (Stella, 2020; Semeraro et al., 2021; Abramski et al., 2023; Basile et al., 2021; Mohammad, 2012), e.g., a text might elicit strong levels of fear and anger, together with lower levels of disgust. Levels might be quantified in terms of how many words elicit those emotions compared to a null model (Stella et

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al., 2020; Mohammad, 2012). The extraction of emotional profiles from text represents a rather recent achievement of AI known as detecting emotional profiles of texts (Mohammad & Kiritchenko, 2015; Semeraro et al., 2022; Stella, 2020; Mohammad, 2012). Several data-informed psychological models map emotions to numerical, vectorial spaces (Posner et al., 2005; Ekman, 1999; Sokolov & Boucsein, 2000), so that emotions can be considered as a superposition of more interpretable dimensions (Sokolov & Boucsein, 2000). For instance, the circumplex model of affect (Posner et al., 2005) represents one of the most widely used models for mapping emotions in a 2D space, extracted from neurocognitive and behavioral data (Gerber et al., 2008): (i) valence, i.e., how pleasant concepts are perceived, and (ii) arousal, i.e., how exciting concepts are perceived. In computer science, valence has often been called ”sentiment”, with the latter term being used to indicate the dimensions of pleasantness/unpleasantness (Liu & Zhang, 2012; Hassani et al., 2020). Other computer science sources tend to use ”sentiment” also to indicate the combination of valence and arousal, creating some confusion (Hassani et al., 2020; Liu & Zhang, 2012). For simplicity, here we use focus on ”valence analysis” to indicate any ”sentiment analysis” considering pleasantness/unpleasantness only. Furthermore, we distinguish between valence analysis and emotional profiling, since emotions need to encapsulate more dimensions than pleasantness/unpleasantness only. In fact, extensive psychological research shows that valence cannot capture the entire range of human emotions (Ekman & Davidson, 1994). For instance, disgust and anger both possess a negative sentiment/valence but disgust inhibits reactions whereas anger creates more alertness and arousal (Ekman, 1999).

The advent of automatic computational analyses promoted emotional text analysis. This resulted in a growing number of studies investigating specific emotions (Bader et al., 2017; Mokryn et al., 2020; Stella et al., 2020; Mohammad & Kiritchenko, 2015; Sulis et al., 2016; Hipson & Mohammad, 2021; Semeraro et al., 2021) and their impact over psychological constructs like depression (Fatima et al., 2021), anxiety (Abramski et al., 2023) and trauma (Abramski et al., 2024), linking emotions to psychometrics. Most computational approaches to emotion detection can be grouped in three clusters (Sarker et al., 2021; Hassani et al., 2020): (i) *lexicon-only* approaches, (ii) *supervised machine learning* approaches, and (iii) neuro-symbolic approaches, merging lexicon encoding, neural processing of information and symbolic parsing. EmoAtlas represents a hybrid approach, falling within the category of neuro-symbolic approaches because it can use AI to detect syntactic and semantic associations between words in text but also perform statistical tests for the presence/absence of emotions in texts via psychologically validated lexicons. Let us focus on the two extremes of lexicon-only and supervised approaches before coming to the functioning of EmoAtlas.

Lexicon-only methods consider texts as collections of words, each one mapping one or more emotions (Mohammad & Turney, 2013). This mapping is contained in a lexicon (Mohammad & Kiritchenko, 2015; Hutto & Gilbert, 2014), i.e., a look-up table where words correspond to specific emotions – e.g., sadness, fear, anger, joy (Mohammad & Turney, 2013) – or norms expressing affective features of words – e.g., valence and arousal (Hutto & Gilbert, 2014). Reconstructing the emotional profile of a text means searching, one by one, all words in the text within the lexicon, providing scores based on word counts (Semeraro et al., 2022) or on pre-computed intensities (Mohammad & Kiritchenko, 2018) as an output. These methods can be computationally inexpensive, as they do not require any machine learning training (Hutto & Gilbert, 2014).

Supervised approaches to text analysis rely on machine learning (Bianchi et al., 2021; Demszky et al., 2020): A classifier is trained to associate each text to a given emotional label. The association is performed through machine learning over features extracted from the text (Abdul-Mageed & Ungar, 2017; Choi et al., 2020; Hassani et al., 2020), like counting the frequency of words (Hassani et al., 2020), or co-occurrences of n words – i.e., n-grams (Dey et al., 2018), or even by transforming words into numerical vectors (Herzig et al., 2017; Pennington et al., 2014), but without performing additional operations other than supervised learning. The output of this training can be a model capable of detecting the binary absence/presence of something, e.g., an emotion, in a given text with a level of confidence (expressed as a probability) (Sprugnoli, 2020; Bianchi et al., 2021; Demszky et al., 2020). An evolution of supervised approaches can be large language models (LLMs), i.e., AIs based on Generative Pre-trained Transformer (GPT) neural architectures (Wang et al., 2024) which use supervised and semi-supervised learning to identify word sequences in human texts and reproduce the latter. In order to be effective, supervised approaches need a considerable amount of textual data for training (Acheampong et al., 2021; Demszky et al., 2020). Pre-trained models can be further trained on specific instances of text corpora to improve performance. However, this extra step comes at a cost, since it requires the addition of extensive manually labeled training data, which is often expensive to collect for various domains (Demszky et al., 2020; Acheampong et al., 2021). Models that do not undergo this additional step are called ”zero-shot” and they are general enough to adjust to almost any topic/jargon being mentioned in texts (Acheampong et al., 2021). The performance of both lexicon-only models and supervised models depends on their inner structure, i.e., the extensiveness of the lexicon (Mohammad & Turney, 2013) or the sample size of training data (Acheampong et al., 2021).

EmoAtlas as a cognitive approach to analyze texts

This work introduces EmoAtlas as a hybrid model performing AI-based syntactic parsing and lexicon-based word-level emotional analysis of texts. Rather than parsing all words co-occurring together in a given sentence, EmoAtlas reconstructs the structure of syntactic/semantic relationships between words in texts. In this way, EmoAtlas links words that are syntactically close but might well appear far apart in a sentence (e.g., "climate" and "bad" in the sentence "climate change is truly, truly, bad"). Unlike co-occurrence networks (Stella, 2022), the syntactic structure obtained through AI parsing reconstructs the associative knowledge embedded by authors in their narratives (Stella, 2020), i.e., how ideas were associated in posts, letters, or books. Such content mapping is performed automatically, exploiting a zero-shot, advanced language model – cf. Honnibal and Montani (2017) – that can parse syntactically large volumes of text without human intervention. Through the extensive emotional lexicon developed by Mohammad and Turney (2013), EmoAtlas counts how many words eliciting emotions were syntactically linked in text to specific targets, thus enabling word-level emotional profiling of texts. Whereas the AI performs syntactic parsing and network construction, the lexicon dealing with emotion detection comes from a psychological mega-study validated by thousands of participants. In this way, EmoAtlas represents a powerful synergy between AI and validated human psychological data.

Testing EmoAtlas on news, posts, and tweets

We test EmoAtlas against different state-of-the-art NLP approaches in performing emotional profiling of news media and social posts in Italian and English. Notice that we also include GPT models like OpenAI's GPT-3.5-0125, Mistral 7B, and LLaMAntino in the comparison to see how performance differs between a model as simple but interpretable as EmoAtlas and systems as powerful but opaque as GPT-based large language models. Notice also that we keep the comparison between EmoAtlas and NLP transformer-based approaches fair, i.e., we do not perform model tuning on any specific dataset. This means that we do not adapt neither EmoAtlas' syntactic parser nor its lexicon to a given specific dataset, and, equally, we do not fine tune any transformer model on a portion of texts from a given corpus. This choice makes it more convenient to test all models even on small-scale datasets and reduces the chance to introduce overfitting issues.

On the emotional profiling of texts: Background literature

In natural language processing (NLP), the emotional profiling of texts can be considered as the natural evolution of valence analysis (Basile et al., 2021). Whereas valence is limited to considering only the positive/negative dimension of language, emotions can capture a wider range of perceptions, like arousal/inhibition or engagement/avoidance mechanisms (Ekman & Davidson, 1994; Ekman, 1999; Plutchik, 2003). Despite the finer grain achievable through emotion detection compared to valence analysis, the latter relatively simpler to implement and easier to interpret (Hassani et al., 2020; Stella, 2022; Mohammad et al., 2017, 2016). For this reason, often some methods were conceived for tackling valence analysis and then extended to emotion detection but within a similar pipeline.

In NLP, there are many different ways to perform emotional analysis, which can be classified into two main classes: (i) methods using text-endogenous features only, either evident or latent, and (ii) methods using text-exogenous features like external lexicons.

Text-endogenous features for emotional profiling

Feature-based emotional profiling of texts transforms them into numerical vectors, turning emotion evaluation into either a classification or a regression problem (Bianchi et al., 2021; Sprugnoli, 2020). These features can be either evident, e.g., interpretable like counting word frequency (Hassani et al., 2020), or latent, e.g., not directly interpretable like a word embedding from a neural network (Abdillah et al., 2020). Interpretable features for emotional analysis include: bag-of-words model, n-gram analysis and word co-occurrence networks. The bag-of-words model is a popular yet simple approach (Hassani et al., 2020): The model represents text as a numerical set of word counts without regard to grammar or word order. This approach has the advantage of being simple to implement and fast to train but it neglect that the meaning of a sentence is not a simple concatenation of the meanings of its constituent words (Mohammad et al., 2016). Bag-of-words models are also sensitive to data sparsity (e.g., texts where many words are used infrequently) (Desmet & Hoste, 2013). A related approach is represented by the analysis of n-grams, i.e., sequences of n consecutive words (Hassani et al., 2020). This approach can capture longer-range dependencies than the bag-of-words model, but is also more computationally expensive, since the word combinations to check can grow exponentially even for short texts (Dakshina & Sridhar, 2014). A third approach is to use co-occurrence networks (Quispe et al., 2021; Silva et al., 2016; Teixeira et al., 2021), which represent words as nodes in a network and connect nodes that co-occur in text. This approach has the advantage of being able to capture the meaning of words in context (Stella et al., 2020) but it suffers from encoding also spurious links between concepts (Teixeira et al., 2021), e.g., connections between words that are not syntactically related in text. EmoAtlas does not produce word co-occurrence networks.

In all the above methods, emotions come from a pre-labeled set, so that a machine learning pipeline can learn to attribute emotional labels to specific sets of features according to training (Demszky et al., 2020; Choi et al., 2020). Latent text-based features can be used for training AI models in analogous ways. Methods like word embeddings, including GLoVe (Pennington et al., 2014) or transformer networks like BERT (Devlin et al., 2018), are also able to transform texts in numerical vectors but such representations cannot be easily interpreted by the experimenter (Rudin, 2019).

Transformer networks (Acheampong et al., 2021; Rudin, 2019; Devlin et al., 2018) are more advanced tools for performing text-endogenous emotional profiling. A transformer network is made up of two parts (Vaswani et al., 2017): (i) an encoder reads the input text and converts it into a sequence of vector representations, while (ii) a decoder takes the vector representations and, through training, is able to make an inference at translating vectors into elements of texts, e.g., tokens. In this way, the transformer can learn statistical relationships between individual words in sentences and approximate their meaning and emotional content based on context. Transformers are designed to handle word-word dependencies through a self-attention mechanism (Vaswani et al., 2017): The network is made to focus on different parts of the input text at different times, filling gaps, completing sentences and producing different vectorial representations of words according to their context (Devlin et al., 2018). Large language models use a transformer architecture in combination with semi-supervised learning to generate more massive training datasets and learn how to produce sequences of words that resemble human texts (Wang et al., 2024).

A crucial downside of BERT (Bidirectional Encoder Representations from Transformers), its related models like RoBERTa, AlBERT, and DistilBERT, and even of more advanced LLMs, is that the embeddings created by attention are mostly opaque to the experimenter (Rogers et al., 2020; Rudin, 2019). To improve model interpretability, novel methods like attention masks (Hao et al., 2021) are being proposed with the aim of understanding how semantic relationships could influence model performance. Given the dependency of these methods over massive training sets, interpreting NLP methods from a cognitive perspective represents a relatively open challenge.

Text-exogenous features for emotional profiling

Lexicon-only approaches to valence analysis represent a valid alternative to the above NLP techniques in several scenarios where computational resources are limited and there is a need for model interpretability over black-box performance (Basile et al., 2021; Stella et al., 2020; Teixeira et al., 2021; Mokryn et al., 2020). These approaches use a set of pre-defined words or phrases (called a lexicon) to identify the emotions of a text. In other words, lexicon-only approaches make use of labeled tables of words, i.e., tables where words are assigned valence scores (Hutto & Gilbert, 2014) or emotional categories (Mohammad & Turney, 2013), to automatically classify the valence or the emotions elicited by a given text. Advantageously, lexicon-only approaches can thus be used with any language for which a lexicon translation is available, while requesting no re-training. Furthermore, lexicons can also be tuned according to the domain or topic of interest – e.g., finance (Loughran & McDonald, 2011) – without needing considerable training like BERT or similar methods require (Rogers et al., 2020). Most lexicon-only approaches do not account for the context in which words are used: a word with a positive valence score in one context may have a negative valence score in another. Another limitation of these models is that the analysis will be as good as the underlying lexicon.

The most powerful lexicons developed in the last few years have been VADER (Hutto & Gilbert, 2014) (which comes as a lexicon and set of rules but is limited to valence only), the Hedonometer (Dodds et al., 2011) (limited to sad/happy emotions), and EmoLex (Mohammad & Turney, 2013), i.e., a lexicon encompassing Plutchik's eight categorical emotions (Plutchik, 2003). VADER (Valence Aware Dictionary for sEntiment Reasoning) is a lexicon method enhanced with rule-based valence analysis tools that is specifically attuned to valences expressed in social media (Hutto & Gilbert, 2014). Thus, VADER has a lexicon of valence scores for words and a set of rules determining how such scores should be combined (e.g., accounting for meaning modifiers in sentences). VADER uses a simple approach to compute the overall valence of a text document by summing up and normalizing the valence scores (i.e., positive, negative, and neutral) of the individual words in the text. Even though it is based on a lexicon (of almost 8000 English words), VADER takes into account both the context and the polarity of the words in a text, e.g., "very good" provides stronger valence than "good". Importantly, VADER also accounts for negations of meanings, emojis, and punctuation. Rather than producing a valence polarity output (positive/negative), VADER computes a valence intensity, e.g., "good" is less positive than "great". In rating the valence of tweets, VADER performed as well as human raters (Hutto & Gilbert, 2014), indicating that even lexicon-only approaches can achieve good performance.

The Hedonometer considers a lexicon of 10,000 English words, gathered from multiple online sources and ranked on a scale from 1 (sad) to 9 (happy) (Dodds et al., 2011, 2015).

In the emotional circumplex of affect model (Posner et al., 2005), happiness roughly corresponds to a state of positive valence and moderate arousal. Valence analysis would not be able to capture the arousal encoded in happiness (Posner et al., 2005). Built through an online survey, the Hedonometer processes 100GB of tweets each day and identifies an average happiness curve as a ”bag-of-word” model (Dodds et al., 2011), e.g., without accounting for syntactic relationships between words in texts. EmoLex (or the National Research Council – NRC – Emotion Lexicon) is a lexicon based on a psychological mega-study, where over 14,000 English words were rated by thousands of human participants (Mohammad & Turney, 2013). Ratings indicated the valence and the emotions elicited by words. Since its original release, almost 10 years ago, EmoLex powered several approaches to emotional analysis, becoming very popular and being used in a wide variety of research projects in psychology, digital humanities, NLP, and public opinion studies (Mohammad & Kiritchenko, 2015, 2018; Semeraro et al., 2021). EmoLex was applied in several tasks to monitor social discourse (e.g., tweets) (Basile et al., 2021; Stella et al., 2020) but also to reconstruct emotional arcs in movies’ narratives (Hipson & Mohammad, 2021) and users’ movie reviews (Mokryn & Ben-Shoshan, 2021; Mokryn et al., 2020).

EmoAtlas as a hybrid approach with two working modes

EmoAtlas combines lexicons and AI-based parsing methods in a hybrid synergy that is uniquely computationally-efficient and interpretable. Figure 1 reports an example showcasing how EmoAtlas performs emotion analysis and builds textual forma mentis networks through AI-based syntactic parsing. EmoAtlas uses AI to identify syntactic relationships between words in texts and can perform either text-level or word-level emotional analysis. In the first case, only a bag-of-words approach is adopted, relying on EmoLex to capture eight emotions in texts (see Methods). In the second case, EmoAtlas combines EmoLex with the network structure to reconstruct the emotional framing of a word by considering its syntactic and semantic associates in text, i.e., those words that were related in text to the target word and thus compose its semantic frame (Semeraro et al., 2022; Stella, 2020).

According to content mapping in the social sciences (Carley, 1994) and frame semantics in psychology (Fillmore & Baker, 2001), it is possible to reconstruct the connotation of a word by checking other ”words that keep company to it” in text: The *semantic frame* of a concept is the set of words associated either syntactically or semantically to that concept within a given (set) of texts (Fillmore & Baker, 2001; Carley,

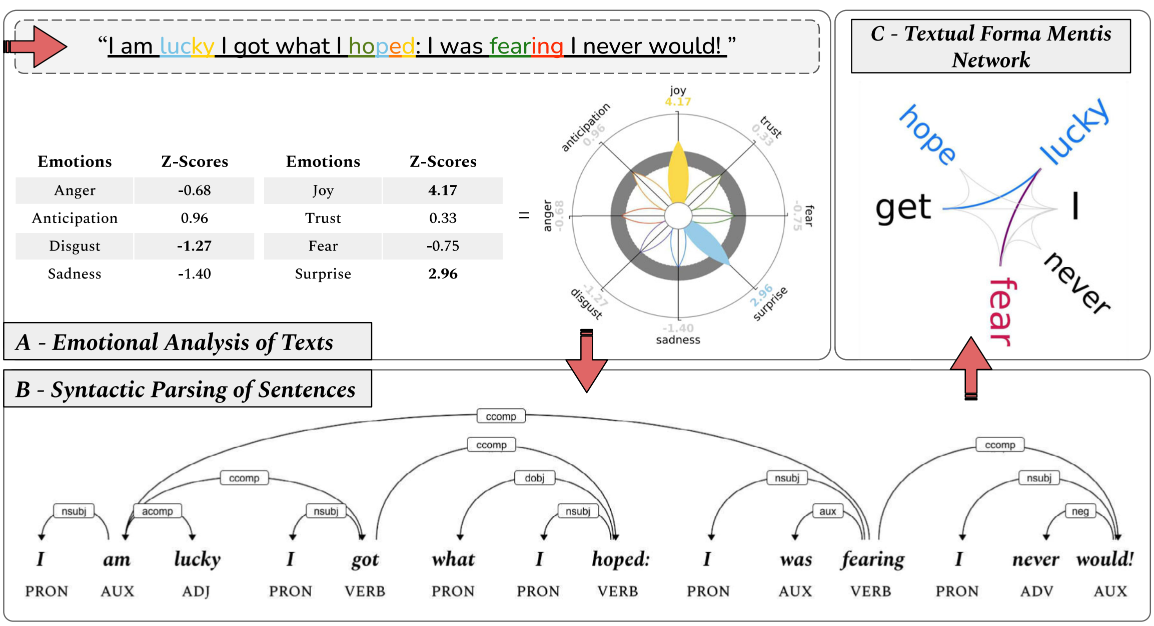

Fig. 1 *Top left*: EmoAtlas reads a text sentence by sentence, mapping words into emotions. The tool can work as in ”bag-of-words” mode, just counting words, or as a network builder. In the ”bag-of-words” mode, words eliciting up to eight emotions are counted and compared against random sampling. Emotional flowers can represent word counts or z-scores. Individual petals indicate scores/z-scores. When using z-scores, colored petals fall outside the rejection region $|z| < 1.96$ ($\alpha = 0.05$), which is indicated by a *grey ring*. *Bottom*: In network-building mode, each sentence goes through syntactic parsing via *spaCy*. On the resulting syntactic tree, the distances t_{ij} separating non-stopwords i and j are measured. Every sampled i and j are linked if $t_{ij} < K$, giving rise to a textual forma mentis networks where: (i) stopwords do not appear, (ii) other concepts are linked despite stopwords not being included. By default $K = 3$, measuring local syntactic relationships on the syntactic tree. *Top right*: Textual forma mentis network of the sentence underlined in (*top*, *left*). *Cyan* (*red*) indicates positive (negative) words as outlined in EmoLex. *Cyan* (*red*) links connect positive words. *Purple* (*gray*) links connect positive and negative (neutral) words

1994). In network-building mode, EmoAtlas can reconstruct the semantic frame of a target word as a network neighborhood of syntactic/semantic associates. EmoAtlas can attribute an emotional profile, i.e., a collection of emotional scores, to a target word by analyzing its semantic frame, i.e., the set of words syntactically specifying or semantically related with the target word as synonyms (e.g., "nature" can also mean "character") or hypernyms/hyponyms (e.g., a "dog" is a type of "animal"). This is performed through the construction of textual forma mentis networks, which identify syntactic/semantic relationships between words in text (Stella, 2020, 2022; Semeraro et al., 2022) (see Methods).

EmoAtlas can perform statistical testings: Rather than providing mere scores like VADER, without a statistical reference, EmoAtlas can quantify how much the emotional words concentrate around a target word, in a given text, when compared against random sampling from a reference corpus. This aspect is crucial to identify the emotional signals that are markedly different from the norm (the reference corpus), rather than simply focusing on emotional signals that tend to be high in general (both in the target text and the reference corpus) (Semeraro et al., 2022).

Thanks to its hybrid AI/lexicon structure, EmoAtlas does not require massive training on huge text corpora, unlike LLMs. This makes it easier to extend EmoAtlas' support to multiple languages besides English. In this work, we show experiments with English and Italian texts, given the expertise of the authors, but showcase and release a technical implementation of EmoAtlas supporting 18 languages: Catalan, Chinese, Danish, Dutch, English, French, German, Greek, Italian, Japanese, Lithuanian, Macedonian, Norwegian, Polish, Portuguese, Romanian, Russian and Spanish.

EmoAtlas and text-as-data approaches in psychometrics

Recovering the emotional/syntactic/semantic content of texts can be valuable for developing novel psychometric approaches powered on texts. Psychometrics deals with the measurement of psychological constructs starting from quantitative data. Whereas most psychometric approaches deal with numerical or categorical ratings, several recent research approaches are starting to extract information from texts in order to perform psychometric estimates. Fatima and colleagues developed and trained an AI in reproducing psychometric scores for anxiety, depression and stress starting from sequences of words in texts (Fatima et al., 2021). Checking the semantic similarity of subsequent words in narratives has been shown to provide valuable insights for measuring creativity levels in texts (Johnson et al., 2022) and assess levels of aphasia severity in clinical populations (Litovsky et al., 2022). Natural language processing vectors were also successfully used to extract personality descriptors made of adjective lists (Cutler & Condon, 2022), whereas syntactic/semantic networks were used to identify expressed distress levels in the narrative of sexual assault survivors using different passive/active voice forms Abramski et al. (2024).

EmoAtlas can be used not only for emotion detection but also for psychometric evaluations of psychological constructs from texts. This task falls within the area of artificial psychometrics, where human-centric artificial intelligence models are developed to measure psychological constructs (Hernández-Orallo, 2017; Johnson et al., 2022).

We test EmoAtlas in both its modes, i.e., "bag-of-words mode" for emotional profiling and "structured" mode for network building, to explore creativity ratings of short stories from (Johnson et al., 2022). Johnson and colleagues collected 1071 five-sentence creative stories, whose creativity was subsequently rated by four human annotators. In that work, BERT outmatched other neural networks and word embeddings in correlating with creativity ratings out of the overall semantic distance between words in a narrative, a measure called divergent semantic integration by the authors (Johnson et al., 2022). Each story contained on average 60 ± 3 tokens.

We trained several AI models to regress either the mean creativity ratings of raters or the scores individually produced by raters (thus creating an ensemble of four AI models, whose estimates could be averaged to match mean ratings), see Methods and results.

Manuscript structure

The manuscript is organized in the following way. The Results section compares EmoAtlas against state-of-the-art transformer networks like BERT, RoBERTa, AlBERT, and DistilBERT when performing three tasks:

- Text-level emotion detection, i.e., assess the presence or absence of a set of emotions in texts, as indicated by human raters. This task is performed in English and in Italian, over a series of different texts, ranging from news headlines (Strapparava & Mihalcea, 2007) to tweets – i.e., the SemEval 2018 Task 1 Tweets dataset (Mohammad et al., 2018) – including online social posts from Reddit (Demszky et al., 2020), YouTube and Facebook (Sprugnoli, 2020).
- Word-level emotion detection and semantic frame analysis, i.e., understanding how reviewers framed their ideas around "beds" in 1-star and 5-star hotel reviews from the Datafiniti's Business Dataset (see https://www.kaggle.com/datasets/datafiniti/hotel-reviews?resource=download, accessed: 10/10/2022).
- Text-based psychometric task, training an AI model to learn emotional and network features of texts and produce creativity ratings for each text being as close as possible

to human ratings. This task is performed in English for 1071 short texts, annotated by four raters (Johnson et al., 2022). This task showcases how text features extracted from EmoAtlas can match BERT scores in evaluating human creativity norms rating short stories.

For the aforementioned tasks, we used BERT, RoBERTa, ALBERT and DistilBERT implementations available on Hugging Face (see https://huggingface.co/bhadresh-savani/bert-base-uncased-emotion, accessed: 09/11/2022), fine-tuned by their authors on the Hugging Face dataset of emotions (Saravia et al., 2018). We also used LLMs like ChatGPT 3.5 in its turbo-0125 version (via API), LLaMAntino 2-Chat-13b-HF-Q8 (in Italian) and Mistral 7B-instruct-v0.2-Q8. These models represent a reasonable choice for out-of-the-box tools that require no further training, but that can tackle the task of emotion annotation.

Across all tasks, the comparison is drawn over all the eight categorical emotions that can be detected by EmoAtlas (some transformer networks support a smaller number of emotions). We test both English and Italian datasets. We also show how the computational cost for running large-scale analyses with EmoAtlas is a mere fraction of the one required by transformer networks, while performance is roughly analogous in most cases.

We discuss our findings in view of relevant literature at the interface of psychology and computer science and provide the links to working code and online tutorials at the end of this manuscript.

Results

Results are organized according to the two languages tested here, English and Italian. We also report on the computational costs of all these routines. We conclude the results section with an outline of the word-level emotional and valence profiling provided by EmoAtlas.

Text-level emotion detection with EmoAtlas in English

Notice that text-level emotion detection is the task of assessing the presence (or absence) of a set of emotions in texts, as indicated by human raters. As reported in the Methods, this is thus a classification task (for each emotion), which can be solved by considering emotional scores or z-scores. Emotional scores identify the absolute presence and intensity of an emotion based on model design, e.g., a confidence level (Acheampong et al., 2021). Emotional z-scores quantify the relative presence and intensity of an emotion compared to a reference null model. See the Methods section for more details on these implementations. The English datasets tested here include tweets from the SemEval 2018 Task 1 Tweets dataset (Mohammad et al., 2018), GoEmotions (Demszky et al., 2020) and news titles (Strapparava & Mihalcea, 2007).

EmoAtlas and tweets

Tables 1, 2 and 3 report accuracy, precision, and recall scores for the eight classification tasks relative to detecting the presence or absence of a given emotion in a tweet from SemEval 2018 Task 1 (Mohammad et al., 2018). Table 1 is relative to emotion detection via emotional scores for BERT-like methods and z-scores for EmoAtlas. Table 2 uses z-scores for both transformers and EmoAtlas, see Methods. Using emotional scores for transformers means binarizing the presence/absence of a given emotion in a given text if the confidence produced by the transformer is higher/lower than a 0.5 threshold. This threshold is the median of the range [0,1], where confidence scores produced by a transformer network belong (Acheampong et al., 2021). EmoAtlas can produce emotional scores too but on a range that varies according to text length. In the absence of a clearly defined upper bound, in EmoAtlas emotional scores cannot be thresholded as easily as in transformer networks. For this reason, we resorted to using statistical thresholds (i.e., z-scores > 1.96) for detecting the presence of an emotion. In Table 1, EmoAtlas was the only viable option to provide results for trust, disgust, and anticipation. These three emotions were not supported by BERT and BERT-like transformers, as they were not trained specifically towards detecting these three emotions. In all other cases, EmoAtlas achieves a slightly lower accuracy compared to BERT and RoBERTa. Table 3 reports results for EmoAtlas against large language models, i.e., ChatGPT 3.5 and Mistral 7B.

Interestingly, precision and recall scores outline a different scenario: EmoAtlas achieves the highest scores in four out of ten comparisons against any of the four BERT-like methods. BERT achieves a way higher precision than EmoAtlas in detecting tweets with fear and surprise but is then outmatched by EmoAtlas in terms of recall. With precision indicating a tendency to avoid producing false positives, this comparison further confirms that the character encoding of BERT (Devlin et al., 2018) is powerful for reducing false positive rates in short texts, compared to lexicon-only approaches, as found also in previous works (Acheampong et al., 2021). However, our results indicate that a good lexicon and the addition of AI syntactic parsing can reduce the occurrence of false negatives, thus leading to instances where EmoAtlas achieves a higher recall than BERT for detecting fear and surprise in short texts.

Always when using emotional scores, ALBERT achieves twice the recall of EmoAtlas and only a marginally lower precision when detecting anger. This is not the case when

Table 1 Model assessments, using emotional scores for transformers and z-scores for EmoAtlas, for binary emotion classification in SemEval 2018 Task 1. ”distb” stands for DistilBERT, ”alb” stands for AlBERT, ”rob.a” stands for RoBERTa, ”emoa” stands for EmoAtlas

Accuracy score %						Precision score %					Recall score %				
	bert	distb	alb	rob.a	emoa	bert	distb	alb	rob.a	emoa	bert	distb	alb	rob.a	emoa
joy	**76.7**	76.1	73.2	75.4	73.7	67.9	67.2	67.1	63.8	**70.2**	67.5	66.6	51.1	**74.1**	47.5
trust	–	–	–	–	**78.5**	–	–	–	–	**9.4**	–	–	–	–	**36.4**
fear	**86.4**	85.1	82.7	84.4	77.3	**71.6**	66.1	58.4	70.3	39.6	41.3	36.4	16.0	24.7	**48.2**
surprise	**94.8**	94.4	94.4	94.7	84.3	**52.9**	38.4	31.7	48.1	7.7	15.2	10.2	4.4	4.4	**18.0**
sadness	75.1	74.3	72.5	**75.6**	71.6	63.0	61.1	63.8	**71.2**	52.3	**36.4**	34.9	14.6	28.6	35.5
disgust	–	–	–	–	**66.8**	–	–	–	–	**73.7**	–	–	–	–	**19.9**
anger	74.3	74.0	69.2	**76.1**	70.0	69.3	68.3	57.1	70.0	**70.1**	55.5	56.2	**68.7**	62.8	33.8
anticipation	–	–	–	–	**74.0**	–	–	–	–	**18.5**	–	–	–	–	**24.0**

Table 2 Model assessments, using emotional z-scores, for binary emotion classification in SemEval 2018 Task 1

Accuracy score %						Precision score %					Recall score %				
	BERT	distb	alb	rob.a	emoa	bert	distb	alb	rob.a	emoa	bert	distb	alb	rob.a	emoa
joy	63.8	63.8	70.8	63.8	**73.6**	0.0	0.0	**73.9**	0.0	70.2	0.0	0.0	29.9	0.0	**47.5**
trust	–	–	–	–	**78.5**	–	–	–	–	**9.4**	–	–	–	–	**36.4**
fear	**86.5**	85.1	82.2	84.5	77.3	**73.4**	67.1	53.4	69.8	39.6	40.1	35.3	17.7	25.8	**48.2**
surprise	94.6	94.3	94.3	**94.7**	85.1	47.0	39.4	33.5	**47.6**	7.7	**18.0**	14.1	8.6	7.7	**18.0**
sadness	74.8	74.2	72.5	**75.2**	71.4	64.4	61.9	63.5	**71.3**	52.3	31.6	31.2	15.0	26.2	**35.5**
disgust	–	–	–	–	**66.8**	–	–	–	–	**73.7**	–	–	–	–	**19.9**
anger	**72.3**	72.0	65.3	69.7	70.0	83.5	83.1	84.6	**88.6**	70.1	32.0	31.3	8.3	21.2	**33.8**
anticipation	–	–	–	–	**74.0**	–	–	–	–	**18.5**	–	–	–	–	**24.0**

Table 3 Model assessments for binary emotion classification in SemEval 2018 Task 1. For large language models, ”ChatGPT” stands for GPT 3.5 125-turbo while ”mistral” stands for Mistral-7B

	Accuracy score %			Precision score %			Recall score %		
	ChatGPT	mistral	emoa	ChatGPT	mistral	emoa	ChatGPT	mistral	emoa
anger	**83.**0	79.7	70.0	78.6	**82.8**	70.2	**74.7**	57.4	33.8
fear	**86.8**	85.4	77.3	62.3	**62.9**	39.6	**68.6**	47.6	48.0
surprise	81.5	**91.9**	85.1	14.7	**26.2**	8.2	**52.4**	29.1	17.7
joy	**84.6**	83.8	73.6	77.1	**85.7**	70.6	**81.9**	66.3	46.6
sadness	**80.9**	78.8	71.4	**76.0**	68.5	52.0	51.1	**51.2**	35.3

Table 4 Model assessments, using emotional scores for transformers and z-scores for EmoAtlas, for binary emotion classification in GoEmotions

Accuracy score %						Precision score %					Recall score %				
	bert	distb	alb	rob.a	emoa	bert	distb	alb	rob.a	emoa	bert	distb	alb	rob.a	emoa
joy	56.4	58.6	63.4	50.6	**73.6**	8.5	8.7	9.3	8.1	**10.9**	75.9	73.1	68.9	**82.1**	56.6
fear	92.7	93.3	94.4	**94.8**	85.0	16.0	17.6	20.0	**21.3**	8.5	42.6	42.6	37.5	36.2	**48.5**
surprise	95.2	95.2	95.5	**96.4**	88.1	26.2	26.0	28.6	**37.8**	7.3	**24.1**	23.7	22.7	11.2	21.9
sadness	86.1	85.4	**91.0**	90.4	85.2	15.7	14.3	16.2	**19.5**	12.7	**40.6**	38.2	19.1	29.5	33.3
disgust	–	–	–	–	**90.5**	–	–	–	–	**11.6**	–	–	–	–	**25.9**
anger	76.2	73.2	70.7	75.1	**85.6**	10.6	9.8	10.1	**11.4**	11.3	52.8	55.3	**64.4**	61.6	28.6

Table 5 Model assessments, using emotional z-scores, for binary emotion classification in GoEmotions

	Accuracy score %					Precision score %					Recall score %				
	bert	distb	alb	rob.a	emoa	bert	distb	alb	rob.a	emoa	bert	distb	alb	rob.a	emoa
joy	**94.8**	**94.8**	**94.8**	**94.8**	73.6	0.0	0.0	0.0	0.0	**10.9**	0.0	0.0	0.0	0.0	**56.6**
fear	92.9	93.3	94.1	**94.6**	85.1	16.6	17.6	19.0	**21.1**	8.5	42.6	42.6	38.4	39.4	**48.5**
surprise	94.8	94.9	95.1	**96.3**	88.5	24.0	24.1	24.9	**38.0**	7.3	**25.9**	24.8	23.7	17.3	21.9
sadness	87.9	87.2	**91.0**	90.7	85.3	17.1	15.2	16.5	**19.9**	12.7	**36.2**	34.1	19.8	28.5	33.3
disgust	–	–	–	–	**90.4**	–	–	–	–	**11.6**	–	–	–	–	**25.9**
anger	89.8	**95.2**	93.6	91.0	85.6	16.5	0.0	9.1	**18.0**	11.3	27.5	0.0	9.0	24.5	**28.6**

z-scores are used instead, where EmoAtlas outmatches ALBERT and all other transformers in terms of recall not only for anger but for seven emotions in the dataset. This finding further indicates that a good lexicon can significantly reduce the occurrence of false negatives for emotional profiling of even short texts like tweets.

The above trade-off and the surprisingly good performance of EmoAtlas in terms of recall when z-scores are applied, added with its support for trust, disgust, and anticipation, indicates that EmoAtlas is competitive with the BERT variants and able to work on more emotions, when z-scores were considered. When using only emotional scores, transformers can provide better results when dealing with anger, whereas EmoAtlas would still provide more accurate results for fear and joy, at a fraction of the computational cost of BERT-like encoders (see the subsection Computational scaling).

For texts as short as tweets, EmoAtlas performs worse than large language models. However, we find zero-shot LLMs perform worse than fine-tuned BERT-like models (see Table 3). Transformers might benefit from token encoding (Vaswani et al., 2017) whereas the counting strategies at work in EmoAtlas might be penalized by shorter text length.

EmoAtlas and GoEmotions

Tables 4, 5 and 6 report accuracy, precision and recall scores for the six classification tasks relative to detecting the presence or absence of a given emotion in a post from GoEmotions (Demszky et al., 2020). Recent work (Chen et al., 2022) showed that the original labeling from GoEmotions, over 27 emotional states, contained several invalid annotations. Since there is no evidence that the errors are systematically biased to favor one method over the others, we preserved the original labeling but considered only those labels equivalent to our emotions rather than mapping all 27 emotional states into the emotions supported by BERT, as done in (Demszky et al., 2020), or EmoAtlas. When using both emotional scores (Table 4) or z-scores (Table 5), accuracy is relatively higher than precision/recall, meaning that the dataset features several true negatives (e.g., neutral posts). Let us thus focus on these two assessment measures. Compared to using emotional scores, z-scores for emotion detection slightly improve model precision but degrade model precision considerably. Hence, let us focus on model assessment using emotional scores (Table 4).

On GoEmotions (Table 4), RoBERTa achieves the highest precision for detecting fear, surprise, sadness, disgust and anger, but is outmatched by EmoAtlas for the detection of joy. RoBERTa does not support disgust detection, which is classified by EmoAtlas with a precision of 11.6%. EmoAtlas outmatches RoBERTa and all other transformers in terms of recall when capturing joy, fear, disgust and anger. BERT slightly outmatches all other models in detecting surprise and sadness. EmoAtlas also outmatches LLMs for detecting anger in terms of precision (Table 6). These results indicate that EmoAtlas on Reddit posts displays a higher tendency to flag more false positives than transformers (lower pre-

Table 6 Model assessments for binary emotion classification in GoEmotions

	Accuracy score %			Precision score %			Recall score %		
	ChatGPT	mistral	emoa	ChatGPT	mistral	emoa	ChatGPT	mistral	emoa
anger	68.8	80.7	**85.6**	11.6	**15.1**	11.1	**82.7**	65.2	28.3
fear	87.9	**89.7**	85.1	12.6	**13.2**	8.5	**61.6**	53.2	48.1
surprise	80.3	**91.6**	88.5	10.2	**16.5**	7.3	**62.2**	37.1	20.9
joy	70.2	**80.8**	73.6	12.9	**16.1**	10.9	**82.6**	63.6	57.1
disgust	73.7	90.2	**90.4**	8.6	**15.0**	11.6	**67.5**	38.0	25.7
sadness	**86.6**	78.4	85.3	**20.9**	14.6	12.8	60.1	**68.4**	33.6

cision, this happens also for LLMs, see Table 6) but it is also less prone to produce false negatives (higher recall). This trade-off, added with the ability for EmoAtlas to surpass transformers in the identification of joy and the support for trust, indicates that LLMs, transformers and EmoAtlas could all be used when analyzing Reddit posts.

EmoAtlas and news titles

Figures 2, 3, and 4 compare models' performance for emotion classification within the headlines dataset (Strapparava & Mihalcea, 2007). Rather than coarse-graining emotional intensities expressed by labelers, we adopted a threshold approach: Emotional intensities higher than a threshold θ indicated the presence of that emotion in that given headline, values strictly lower than θ denoted a lack of emotion, instead. This binarization led to multiple classification tasks, one per each value of θ selected. In the top 6 panels of Figs. 2 and 3 emotional scores were used for transformer networks,

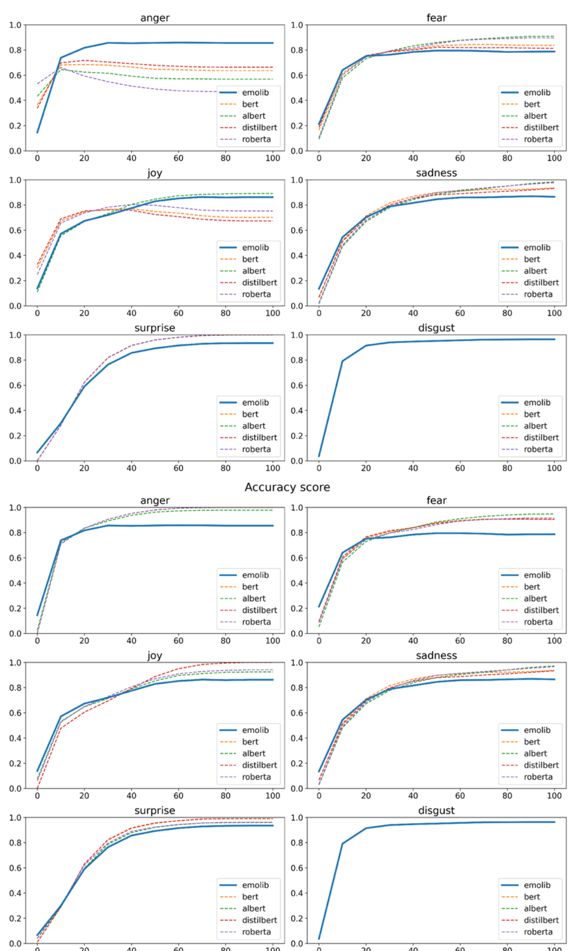

Fig. 2 Accuracy scores, for different thresholds, of emotion classification for the English news dataset, comparing EmoAtlas, BERT, RoBERTa, AlBERT and DistilBERT. *Top*: Model performance when scores are used. *Bottom*: Model performance when z-scores are used

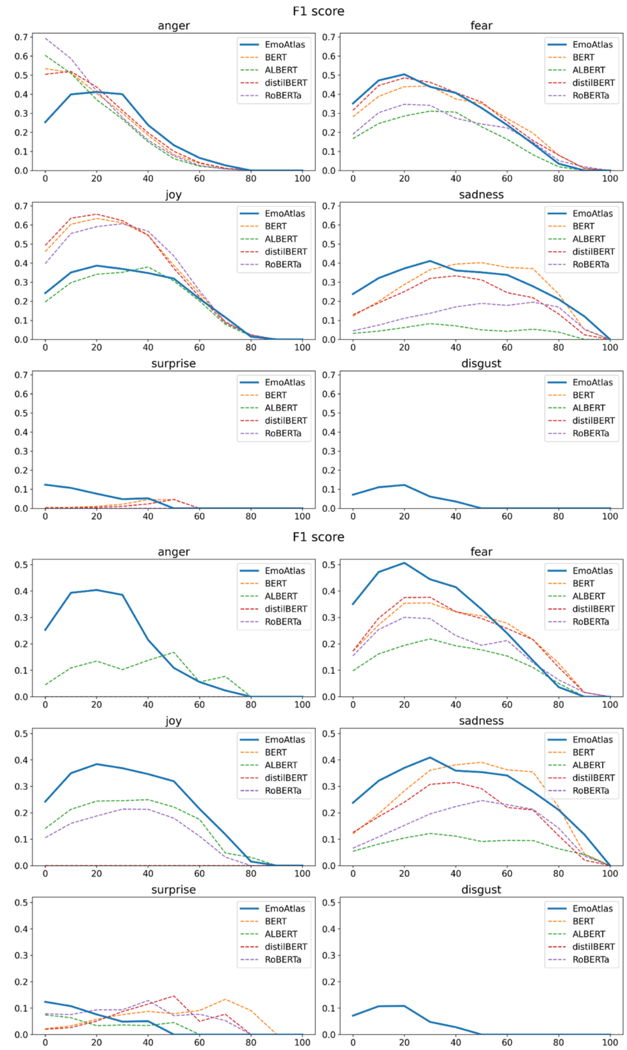

Fig. 3 F-1 scores, for different thresholds, of emotion classification for the English news dataset, comparing EmoAtlas, BERT, RoBERTa, ALBERT and DistilBERT. *Top*: Model performance when scores are used. *Bottom*: Model performance when z-scores are used

whereas z-scores were used for EmoAtlas. In the bottom six panels of Figs. 2 and 3 and in all panels of Fig. 4, z-scores were used for all methodologies (see Methods). To condense information, rather than analyzing precision and recall, we show its harmonic mean, i.e., the F1-score.

When using scores for transformers, EmoAtlas achieves similar values to all other BERT-like transformers in detecting fear and sadness. EmoAtlas performed as badly as ALBERT and achieved only half the F1 score as BERT, DistilBERT, and RoBERTa in identifying joy. Interestingly, EmoAtlas performed slightly better than all other BERT-like transformers in detecting stronger anger, surprise, and disgust. Transformers behaved quite badly in capturing surprise and disgust, with F1 scores close to 0 for all values of Θ. Interestingly, EmoAtlas achieves accuracies analogous to LLMs. When detecting anger, fear, joy, surprise, and disgust,

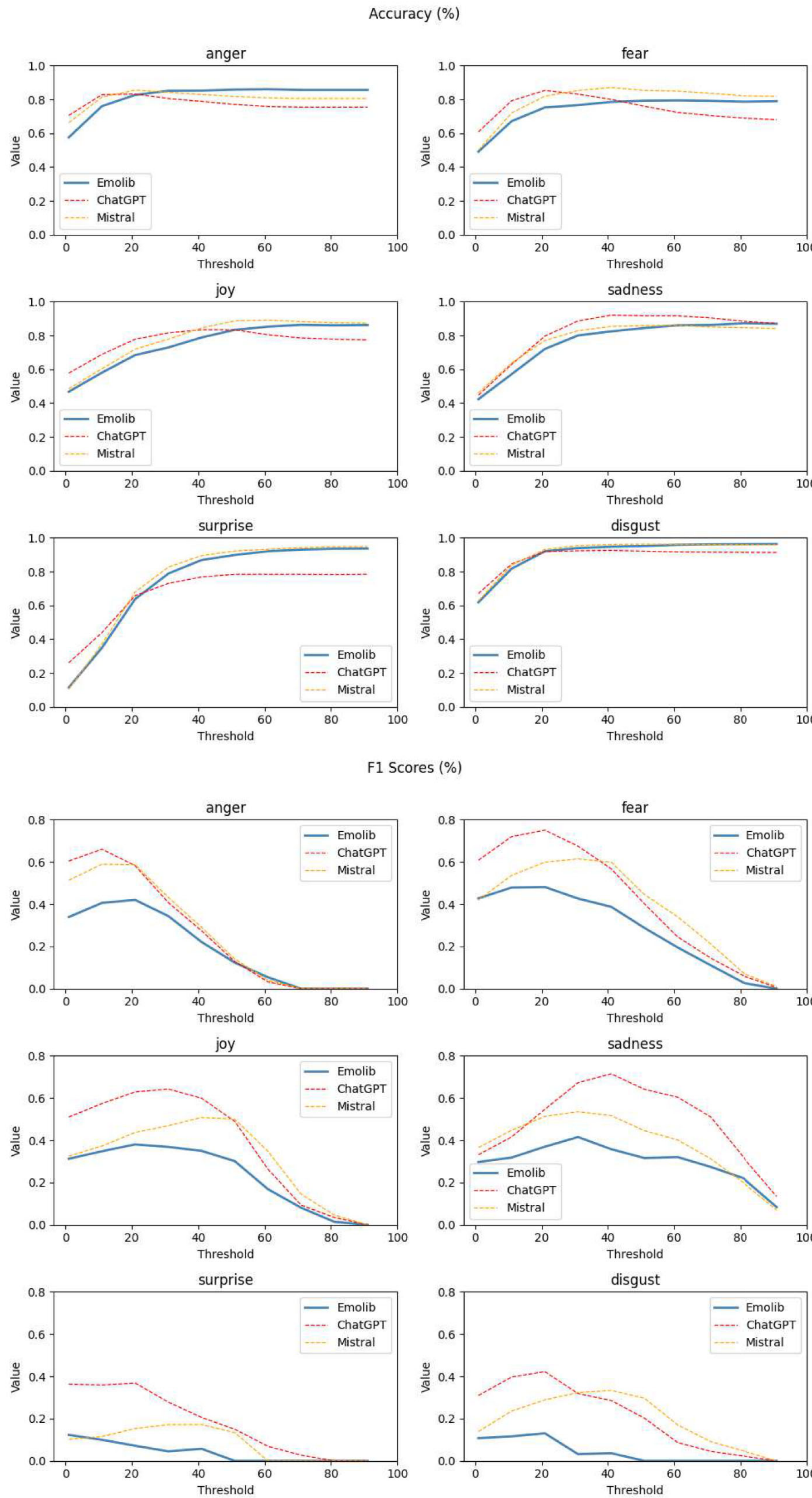

Fig. 4 Accuracy scores (*top*) and F-1 scores (*bottom*), for different thresholds, of emotion classification for the English news dataset, comparing EmoAtlas and LLMs when *z*-scores are used

EmoAtlas can perform better than ChatGPT or Mistral. However, this comes at the cost of EmoAtlas achieving lower F-1 scores compared to LLMs (cf. Fig. 4).

These findings indicate that out-of-box BERT-like networks should not be used for detecting disgust or surprise in news headlines. These emotions would be challenging to detect even for EmoAtlas; however, it would still perform better than transformers. This difficulty was observed also in the original challenge (Strapparava & Mihalcea, 2007) and it might be due to headlines using jargon that encodes surprise and disgust in different ways compared to common language. Since both such emotions can inhibit attention and lower engagement, it is expected for news headlines to attract attention and engagement in different ways when encoding such emotions, ways that are more difficult to be interpreted by transformers and EmoAtlas. Importantly, the above comparison indicates that EmoAtlas produces equivalent, if not slightly better results, when detecting stronger levels of anger, fear, and sadness in headlines compared to transformers, but at a mere fraction of the computational costs of the latter (see Computational scaling).

EmoAtlas in Italian

The English datasets tested here include MultiEmotions-it (Sprugnoli, 2020) and FEEL-it (Bianchi et al., 2021).

EmoAtlas and MultiEmotions-it

Tables 7 and 8 report accuracy, precision and recall scores for the classification of eight emotions in an online post (Facebook posts and YouTube comments) from MultiEmotions-it in Italian (Sprugnoli, 2020). Accuracies are way higher than precision and recall scores, indicating that the dataset contains several true negatives and that accuracy is thus an unreliable estimator for assessing model performance.

Let us focus on Table 7, when emotional scores (z-scores) are used for transformers (EmoAtlas). In terms of precision, EmoAtlas outmatches transformer networks in detecting four out of the five emotions available to both EmoAtlas and BERT-like models. BERT achieves a slightly higher precision for the classification of sadness (31% vs. 27% by EmoAtlas). In terms of recall, EmoAtlas surpasses transformer networks in detecting six out of eight emotions, except for joy and anger, where DistilBERT and AlBERT are the best models, respectively. Analogous results hold also when z-scores are used for classification. The positive performance of EmoAtlas might be motivated by a powerful lexicon available for the Italian language. As reported in Table 9, EmoAtlas outperforms in terms of accuracy GPT 3.5 and LLaMAntino in Italian for the detection of surprise, joy, and disgust, although all models suffer from relatively lower precision scores.

The above results indicate that in Italian, the approach of EmoAtlas is as powerful as Italian-supporting transformer architectures in detecting a wide range of categorical emotions, including trust, fear, sadness, disgust and anticipation, in comments posted on online platforms like Facebook and YouTube.

EmoAtlas and FEEL-it

Tables 10 and 11 report accuracy, precision and recall scores for the classification of four emotions in Italian tweets from FEEL-it (Bianchi et al., 2021). This dataset labeled only such four emotions, i.e., fear, joy, sadness, and anger.

Over this dataset, results are mixed. EmoAtlas achieves the highest accuracy and precision scores for detecting joy but it is then outmatched by DistilBERT in terms of recall. DistilBERT and RoBERTa identify fear and sadness with higher precision than EmoAtlas but both transformer networks are vastly outmatched by EmoAtlas in terms of recall.

When using emotional scores, transformer networks achieved very low recall values, i.e., extremely high false negative rates, for detecting fear and sadness in Italian tweets. This indicates a potential issue with the training of these mod-

Table 7 Model assessments, using emotional scores for transformers and z-scores for EmoAtlas, for binary emotion classification in MultiEmotions-it

	Accuracy score %					Precision score %					Recall score %				
	bert	distb	alb	rob.a	emoa	bert	distb	alb	rob.a	emoa	bert	distb	alb	rob.a	emoa
joy	53.7	40.8	**72.4**	54.7	62.0	23.2	23.0	21.7	26.4	**28.3**	46.6	**71.5**	9.8	58.5	48.3
trust	–	–	–	–	**55.3**	–	–	–	–	**75.2**	–	–	–	–	**31.1**
fear	97.2	97.3	**97.9**	97.8	93.1	0.0	0.0	0.0	0.0	**6.4**	0.0	0.0	0.0	0.0	**18.4**
surprise	**84.2**	**84.2**	**84.2**	**84.2**	71.7	0.0	0.0	0.0	0.0	**8.3**	0.0	0.0	0.0	0.0	**7.8**
sadness	**78.5**	**78.5**	**78.5**	78.3	71.4	**31.0**	20.0	0.0	18.1	27.0	0.4	0.3	0.0	0.3	**20.2**
disgust	–	–	–	–	**91.2**	–	–	–	–	**30.1**	–	–	–	–	**15.8**
anger	56.8	78.8	26.2	62.9	**86.2**	12.4	18.4	11.7	13.6	**23.1**	48.3	26.4	**87.2**	44.2	10.6
anticipation	–	–	–	–	**70.2**	–	–	–	–	**5.0**	–	–	–	–	**37.0**

Table 8 Model assessments, using emotional z-scores, for binary emotion classification in MultiEmotions-it

	Accuracy score %					Precision score %					Recall score %				
	bert	distb	alb	rob.a	emoa	bert	distb	alb	rob.a	emoa	bert	distb	alb	rob.a	emoa
joy	77.6	**77.8**	73.2	76.8	62.0	22.1	0.0	22.0	**34.1**	28.3	3.2	0.0	8.7	7.3	**48.3**
trust	–	–	–	–	**55.3**	–	–	–	–	**75.2**	–	–	–	–	**31.1**
fear	**93.9**	93.3	91.3	92.5	93.1	1.1	2.7	2.7	2.6	**6.4**	5.6	4.9	11.5	8.7	**18.4**
surprise	**83.0**	80.7	81.0	79.0	71.7	13.9	2.9	9.1	**18.8**	8.3	1.6	1.4	1.8	**10.3**	7.8
sadness	76.5	77.1	76.5	**77.9**	71.4	20.8	17.8	9.6	25.0	**27.0**	2.6	1.6	1.4	1.6	**20.2**
disgust	–	–	–	–	**91.2**	–	–	–	–	**30.1**	–	–	–	–	**15.8**
anger	86.9	83.9	**89.0**	84.8	86.2	18.3	19.5	0.0	16.9	**23.1**	5.5	**14.7**	0.0	10.3	10.6
anticipation	–	–	–	–	**70.2**	–	–	–	–	**5.0**	–	–	–	–	**37.0**

Table 9 Model assessments for binary emotion classification in MultiEmotions-it

	Accuracy score %			Precision score %			Recall score %		
	ChatGPT	LLaMAntino	emoa	ChatGPT	LLaMAntino	emoa	ChatGPT	LLaMAntino	emoa
anger	86.2	**89.4**	86.6	42.9	**52.7**	23.1	**75.4**	40.8	9.2
fear	**95.0**	87.7	93.8	**13.7**	7.0	6.5	27.3	**40.9**	15.2
surprise	69.8	30.5	**72.5**	16.1	**16.3**	8.9	21.6	**81.9**	8.0
joy	60.0	23.9	**60.2**	**34.8**	21.7	28.1	92.6	**93.3**	51.3
disgust	83.4	84.9	**91.2**	26.5	19.3	**28.7**	**72.3**	34.0	14.0
trust	**62.4**	42.5	54.8	74.4	47.4	**75.4**	**51.1**	14.8	29.8
sadness	**81.4**	74.0	71.6	**61.4**	28.4	26.5	**35.4**	14.1	18.3

Table 10 Model assessments, using emotional scores for transformers and z-scores for EmoAtlas, for binary emotion classification in FEEL-it

	Accuracy score %					Precision score %					Recall score %				
	bert	distb	alb	rob.a	emoa	bert	distb	alb	rob.a	emoa	bert	distb	alb	rob.a	emoa
joy	54.6	49.8	64.9	62.4	**70.5**	41.5	39.9	56.6	48.2	**61.7**	65.0	**80.3**	7.4	66.0	46.3
fear	94.6	94.6	**94.9**	**94.9**	87.2	0.0	**22.2**	0.0	0.0	16.3	0.0	4.3	0.0	0.0	**37.2**
sadness	85.4	85.4	**85.5**	**85.5**	76.0	32.5	0.0	0.0	**46.4**	23.4	1.1	0.0	0.0	3.3	**28.7**
anger	57.8	60.0	48.5	**61.1**	59.1	54.6	67.2	46.3	57.4	**67.6**	39.0	20.6	**93.2**	47.2	16.6

Table 11 Model assessments, using emotional z-scores, for binary emotion classification in FEEL-it

	Accuracy score %					Precision score %					Recall score %				
	bert	distb	alb	rob.a	emoa	bert	distb	alb	rob.a	emoa	bert	distb	alb	rob.a	emoa
joy	64.3	64.3	65.4	67.3	**70.5**	0.0	0.0	58.4	**88.2**	61.7	0.0	0.0	10.5	10.3	**46.3**
fear	**93.0**	91.7	89.7	91.2	87.2	13.0	12.5	7.2	12.4	**16.3**	6.8	11.0	8.7	11.2	**37.2**
sadness	83.6	83.7	82.9	**84.9**	76.0	14.3	18.5	14.8	**33.9**	23.4	2.7	3.0	4.6	4.4	**28.7**
anger	58.2	58.8	55.2	57.9	**59.1**	70.0	**71.9**	0.0	68.6	67.6	12.7	12.8	0.0	10.5	**16.6**

Table 12 Model assessments for binary emotion classification in FEEL-it

	Accuracy score %			Precision score %			Recall score %		
	ChatGPT	LLaMAntino	emoa	ChatGPT	LLaMAntino	emoa	ChatGPT	LLaMAntino	emoa
anger	**88.6**	73.1	59.1	87.0	**89.7**	67.9	**87.6**	45.1	16.1
fear	**90.7**	68.4	87.2	**32.4**	8.1	18.3	**76.7**	50.5	40.8
joy	**93.3**	36.0	70.5	**90.2**	35.4	62.0	91.0	**96.4**	48.0
sadness	**89.1**	68.8	76.0	**61.8**	18.0	23.9	**64.2**	32.8	30.4

els in Italian. The situation improved slightly when using z-scores, when transformer networks achieved recall scores between 0 and 12.8% across all four emotions. In the same setting, EmoAtlas achieved recall scores significantly higher, between 16.6% and 46.3%. Transformer networks achieved marginally better precision scores than EmoAtlas (i.e., 3 or 4 points above) for sadness and anger. RoBERTa outperformed EmoAtlas in terms of precision for detecting joy but still presented an issue with model recall, where EmoAtlas obtained four times the recall obtained by RoBERTa.

The above results indicate that detecting emotions in texts as short as tweets with out-of-box transformers in Italian was a challenge for BERT-like architectures (in terms of mediocre recall scores). The situation improves substantially with LLMs, which achieve considerably better precision and recall than EmoAtlas (cf. Table 12). Despite the challenge of considering short texts, EmoAtlas might represent a better solution compared to transformers, especially in terms of employing a model with moderate precision but significantly better recall rates.

Computational scaling

The above results provide a richly complex picture where EmoAtlas is not an absolute winner but can still outmatch BERT-like transformer architectures in the emotional profiling of many types of texts, including tweets, Facebook posts, YouTube comments, and news headlines. This is a remarkable achievement for a hybrid model that is not trained on, and cannot thus adapt to, text data. In several instances, especially with shorter texts, transformer architectures achieved only slightly superior recall and precision scores, with differences around 3%. At the same time, LLMs outperformed EmoAtlas by an average of 4 points in terms of accuracy in most instances. The divide got more significant for shorter texts, e.g., tweets, but EmoAtlas is considerably less complex than most LLM architectures, whose training is expensive and, if based on APIs, also costly. Because of these differences, a more appropriate comparison would be the one between EmoAtlas and BERT-like models. In such a complex interplay, where EmoAtlas behaves analogously to transformer networks, what are the computational costs for these implementations of emotional analysis?

Figure 5 reports the execution time in seconds of emotional analysis instances producing emotional scores on artificial texts (i.e., collections of X tweets). For both EmoAtlas and transformer architectures, longer texts (in terms of more tweets to analyze) require linearly longer execution times. This is expected: all models have to work serially in processing a growing number of texts/tweets. DistilBERT produces emotional profiles in less than 40% of the time required by ALBERT and BERT, which is expected (Sanh et al., 2019). This means that DistilBERT is $1/0.4 = 2.5$ times faster than BERT in performing emotional analysis. EmoAtlas is 12 times faster than BERT in achieving the same task of emotional profiling of the same set of tweets.

Figure 6 indicates that whereas parsing 2500 tweets might take more than 5 min with BERT (two with DistilBERT), it might take only 18 s with EmoAtlas. Longer texts might improve this considerable time difference between transformer networks and EmoAtlas. Combined with the above results of relatively very good performance for EmoAtlas, this plot indicates that EmoAtlas is a lightweight and convenient model for investigating emotions in large corpora,

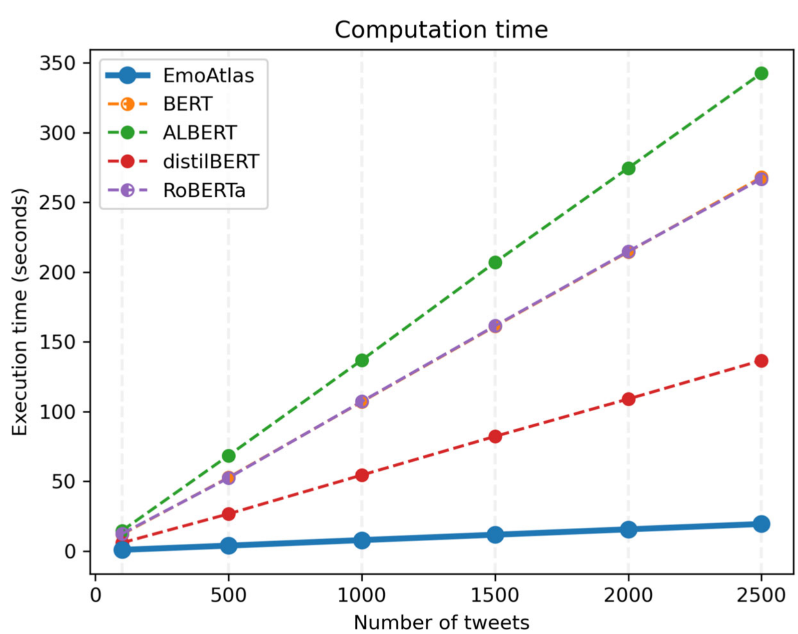

Fig. 5 Computational time for emotional profiling of texts of growing length (i.e., number of tokens) for BERT, BERT-like transformers and EmoAtlas

Fig. 6 *Top*: Network neighborhoods/semantic frames for "bed" in 100 5-star reviews (*left*) and 100 1-star reviews (*right*). *Cyan* (*red*) indicates positive (negative) words as outlined in EmoLex. *Cyan* (*red*) links connect positive words. *Purple* (*gray*) links connect positive and negative (neutral) words. *Bottom*: Emotional flowers for "bed" in 100 5-star reviews (*left*) and 100 1-star reviews (*right*). Petals indicate z-scores. Colored petals fall outside the rejection region $|z| < 1.96$ ($\alpha = 0.05$) as indicated by a beige ring (whose width indicates the rejection region itself). Petals below (above) the ring indicate a text way poorer (richer) than the baseline in emotional content. Z-scores falling in the rejection region are greyed out. Reviews taken from the Datafiniti's Business Dataset (see https://www.kaggle.com/datasets/datafiniti/hotel-reviews?resource=download, accessed: 10/10/2022)

achieving results comparable with way slower transformer architectures.

Our above results indicate that EmoAtlas has a considerable speed advantage over BERT-like classifiers and also supports more emotions. This positive side is balanced by a cost of losing around 3% in terms of accuracy. LLMs perform only marginally better than BERT-like classifiers. However, large language models currently require either expensive computational setups (e.g., high-end GPUs) or API costs for running automated estimations. Even in the presence of high-end GPUs, the time for reading texts longer than a few sentences can fluctuate widely, based on whether the model is loaded on the GPU only or it requires additional RAM. Ultimately, these elements translate into a relative difficulty in estimating computational times for LLMs and make the latter less accessible than EmoAtlas to individual researchers without high-end hardware setups. EmoAtlas can currently run on services like Google Colab with no installation or computational requirement, which is an additional benefit that should be mentioned when considering computational efficiency.

Achieving interpretable semantic frame analysis with EmoAtlas

Transformer networks, BERT-like and LLMs, are mostly black box models and their understanding of the emotional structure of text cannot be easily directly understood by researchers yet (Rudin, 2019). Instead, EmoAtlas provides advanced quantitative tools, relying on cognitive network science and psychological theories, for better understanding and interpreting the syntactic, semantic and emotional structure of associations between words in texts.

Word-level emotional analysis denotes a rapidly developing sub-field of affective computing research where one wants to identify the emotions attributed to a specific concept in a text, i.e., a word or a name (Hassani et al., 2020). Notice that this is different from entity-level emotional analysis, where focus could be given to entities being represented also by two or more words, e.g., "Santa Claus" (Hassani et al., 2020; Abramski et al., 2024). Whereas it is relatively easy to deconstruct a text in its words, entity identification remains a challenging problem for which machine learning or other advanced techniques are required (Hassani et al., 2020). For instance, Semeraro and colleagues (Semeraro et al., 2022) reconstructed a positive, trustful perception of the concept of COVID-19 "vaccines" as reported in 3000 Italian news from mainstream journals. The same concept was framed in neutral terms in a similarly sized corpus of Italian news produced by alternative sources.

Figure 6 reports the semantic frame for "staff" in 100 5-star reviews (left) and 100 1-star reviews (right) from the Datafiniti's Business Dataset (top). The data visualizations report semantic frames as a network of syntactically/semantically linked concepts and as emotional flowers. The network representation provides insights about how words are syntactically/semantically related across all sentences composing all text in the investigated corpus, e.g., all 1-star reviews. The petals showcase which emotions are stronger (weaker) than a null model in the considered semantic frame, e.g., collating all network neighbors. We refer the reader to the Methods section (and the caption of Fig. 6) for more details. As mentioned in the Introduction, semantic frames are built as neighborhoods of textual forma mentis networks, i.e., network neighborhoods of networks detecting syntactic/semantic relationships between words in texts (Stella, 2020, 2022). The bottom of Fig. 6 displays the emotional profiles for "bed" in such texts. Reviews differ in terms of self-reported customer experience. A 5-star experience will be narrated as a fantastic experience, potentially magnifying positive features of "bed" and referencing it along positive emotional jargon (e.g., "the bed was spacious and comfy!"). Instead, a 1-star experience will be narrated as a dreadful experience, potentially outlining "bed" along a negative connotation, with jargon eliciting negative emotions ("the bed was full of nasty bugs!"). This is reflected in the examples reported here, relative to the first 100 reviews in the dataset for each star rating.

In five-star reviews, "bed" was framed mostly through positive/neutral associations, i.e., blue/gray links in Fig. 6 (top, left). The syntactic associates of "bed" in 5-star reviews constitute a semantic frame populated by strong levels of joy and trust, stronger than random expectation, cf. Figure 6, bottom left. These positive emotions vanished in the semantic frame for the same object, "bed", in 1-star reviews. Interestingly, the 1-star semantic frame features several negative associations, including mentions of bed bugs and poor hygiene practice.

This example showcases how EmoAtlas can be used to track specific connotations of individual words/concepts in different corpora, tracking not only the emotions populating semantic frames but also their content in terms of word associations. EmoAtlas can thus provide a data-driven measurement of contextual valence shifting (i.e., words changing their valence according to the contexts they feature in, like "bed" in Fig. 6) or historical evolution (Turney & Mohammad, 2019) (i.e., words changing their valence and emotions according to historical contexts). In this way, EmoAtlas enables quick and easy quantitative experiments reconstructing semantic frames, connotations, and emotional profiles of words across hundreds of texts with relatively little effort.

EmoAtlas and quantitative evaluations of psychological constructs

Here we test how EmoAtlas fares against BERT in reproducing the mean creativity ratings of short stories, collected and rated in (Johnson et al., 2022). We here limit our analysis to BERT in order to adhere to the original study by Johnson and colleagues. In the dataset by Johnson et al. (2022) there are 1071 stories, each one rated independently by four human raters on a creativity scale from 1 (low creativity) to 5 (high creativity). These rates constitute gold estimates for the creativity perceived in short stories. For each short story, we extract its emotional profile (in terms of eight z-scores) and textual forma mentis network. Getting inspiration from previous works using network features to investigate creativity levels (Stella & Kenett, 2019; Kenett et al., 2018), we transform TFMNs into numerical features by measuring their: (i) average network distance between words, (ii) mean local clustering between words, and (iii) inverse network diameter. These measures consider how well connected words are in terms of their syntactic relationships. We expect for texts giving rise to less clustered and less connected networks to correspond with more loosely syntactically/semantically related narratives, where more remote/longer paths connect any two concepts. Following Mednick's theory of creativity (Mednick, 1962) that more remote associations correspond to higher creativity, we expect for more loosely connected narratives/networks to have higher creativity ratings. A larger distance between words corresponding with higher creativity levels represents the same working hypothesis of the divergent semantic integration (DSI) score presented in (Johnson

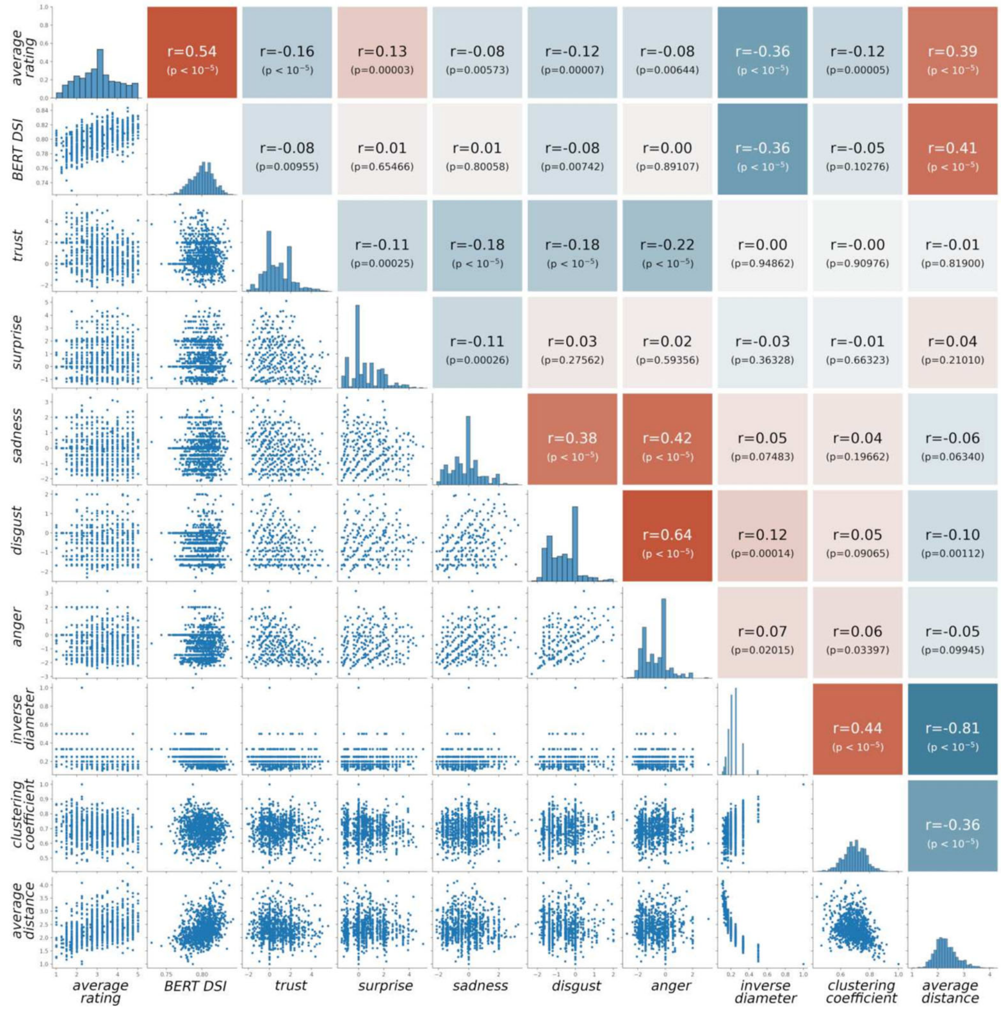

Fig. 7 Pearson's correlations, distributions, and scatter plots of the features used here for investigating average human creativity ratings ('average rating') of 1071 short stories (data from (Johnson et al., 2022). Positive (negative) correlations are highlighted in *light red* (*blue*). The main diagonal contains histogram plots for every considered network feature: mean creativity ratings (Johnson et al., 2022), BERT DSI scores (Johnson et al., 2022), emotional *z*-scores (from EmoAtlas), inverse network diameter (Newman, 2018), mean local clustering coefficient (Newman, 2018), average network distance between nodes in the largest connected component (Newman, 2018) (all three from EmoAtlas). The lower diagonal contains scatter plots between these quantities

et al., 2022), but from a semantic, rather than syntactic, perspective.

Figure 7 reports a summary of the emotional/network features of short stories extracted by EmoAtlas, together with the average human ratings of creativity and BERT's estimations of DSI scores. Figure 7 contains a summary of all correlations between features and the target average creativity rating (i.e., average rating). The latter is the dependent variable we want to train our AI on. The independent variables we attribute to each story are: BERT DSI scores, from (Johnson et al., 2022), emotional z-scores (from EmoAtlas) and three network-based features (always extracted from stories as parsed by EmoAtlas). BERT's DSI scores were obtained by summing all cosine distances between word embeddings produced by BERT while parsing each story (Johnson et al., 2022). BERT's DSI scores correlate positively with creativity ratings (Pearson's $\rho = 0.54$, $N = 1071$, $p < 10^{-5}$, see Fig. 7): Higher scores correspond with higher similarity ratings, confirming the assumption that narratives with more semantically unrelated words are rated more highly in terms of creativity. Whereas BERT's DSI scores are mostly based on semantic representations, the TFMNs extracted by EmoAtlas are mostly based on syntactic relationships between concepts. The average distance of concepts in the largest connected components (Newman, 2018) of these syntactic networks correlates positively both with BERT's scores (Pearson's $\rho = 0.42$, $N = 1071$, $p < 10^{-5}$) and human ratings (Pearson's $\rho = 0.42$, $N = 1071$, $p < 10^{-5}$). This finding indicates that larger average syntactic distances between words tend to correspond to higher ratings. Interestingly, the inverse diameter and the local clustering correlate negatively with human ratings (see Fig. 7): Narratives where syntactic links tend to cluster less and improve the maximum distance between nodes tend to be rated as more creative. These three findings indicate that a phenomenon of divergent syntactic integration might co-exist together with the DSI.

Importantly, Fig. 7 highlights that the z-scores for five out of eight basic emotions, as implemented in EmoAtlas, correlate with human ratings (at a significance level of 0.05). Lower emotional intensities (z-score) for emotions like trust, surprise, anger, disgust, and sadness, correlate with higher creativity ratings, suggesting that creativity might arise as a construct more heavily influenced by intelligence-focused associations rather than affective ones, as postulated already by Mednick (1962).

We implemented all the features reported in Fig. 7 in a supervised regression task. Using a fourfold cross-validation, we trained a random forest to learn and predict human creativity ratings of short stories based on their emotional, network and BERT-based features. Having tested linear regression, multilayer perceptrons and random forests, as implemented in Mathematica 11.3, we found that random forests achieved the highest correlations between predictions and empirical test values.

Table 13 reports model assessment for different sets of features (i.e., models) predicting human creativity ratings of 1071 short stories. Interestingly, the emotional profiles provided by EmoAtlas are enough to train an AI providing creativity estimates correlated with the human ones ($\hat{\rho} = 0.345$, $N = 268$, $p < 10^{-3}$). In other words, the emotional profiles identified by EmoAtlas convey useful information about the estimated creativity levels of short stories. Syntactic network features achieve a higher prediction power ($\hat{\rho} = 0.480$, $N = 268$, $p < 10^{-4}$), i.e., a stronger correlation between predicted and human ratings of creativity. This indicates that the network-building mode of EmoAtlas can provide insightful measurements for the investigation of psychological constructs based on texts.

Table 13 also indicates that BERT's DSI scores are slightly superior to network features in predicting human creativity ratings of short stories. Noticeably, BERT and EmoAtlas achieve equivalent performance in estimating creativity scores, around a Pearson's correlation $\hat{\rho} \approx 0.495$ ($p < 10^{-3}$). This represents quantitative evidence that next to a semantic divergence of concepts, there is also a syntactic divergence and an emotional connotation of short stories: Unveiling and combining the latter two through EmoAtlas can give predictions of the same quality of BERT, which focuses on semantics but requires considerably higher computational costs (see Computational Scaling). BERT and EmoAtlas providing equivalent results and BERT being the best model for estimating DSI (Johnson et al., 2022) both stress the fact that EmoAtlas can be a powerful tool for artificial psychometrics (Hernández-Orallo, 2017), i.e., the quantitative investigation of psychological constructs through artificial intelligence.

Table 13 Pearson's correlation $\hat{\rho}$, p values and root mean square errors (RMSE) for random forest models based on different features: (i) the emotional z-scores of trust, surprise, anger, disgust, and sadness (emotions only), (ii) inverse network diameter, mean local clustering coefficient and average network distance between words in the same largest connected component (network features only), (iii) BERT's DSI scores (BERT only), (iv) emotional z-scores and network features (emotions & network), (v) BERT's DSI scores, emotional z-scores and network features (BERT & emotions & network)

Model	Pearson's $\hat{\rho}$	p value	RMSE
Emotions only	0.345	$< 10^{-3}$	0.895
Network features only	0.480	$< 10^{-4}$	0.841
BERT only	0.494	$< 10^{-4}$	0.861
Emotions & network	0.495	$< 10^{-4}$	0.822
BERT & emotions & network	0.603	$< 10^{-4}$	0.759

Across a fourfold cross-validation, for each model/set of features, we only reported the results for the folds with the lowest correlations. All correlations $\hat{\rho}$ are relative to predicted/artificial and empirical/human creativity ratings over the test set

Since BERT focuses on the semantic aspects of words while EmoAtlas focuses on syntactic and emotional relationships, one could assume that BERT's scores and EmoAtlas' features might be combined to produce a better predictor. Combining these features leads to better performance: A model capturing human mean creativity ratings of stories with a correlation of $\hat{\rho} \approx 0.603$ ($p < 10^{-4}$) through only nine features, eight of them being interpretable (i.e., understandable as measures of network structure or statistical comparisons for word counts). Rather than training one AI on mean creativity ratings, we also trained four AIs on the ratings provided by the four raters of the original study (Johnson et al., 2022). Averaging the predictions of these four regressors led to an even higher correlation with the average empirical creativity ratings, $\hat{\rho} \approx 0.628$ ($p < 10^{-4}$). Our approach is thus psychometric in the sense that it can be applied to ratings produced by individuals, as well as to means of ratings. In both cases, the boosts in performance confirm that BERT and EmoAtlas can provide different information about texts and could thus be used in synergy for the exploration of psychological constructs in texts.

Discussion

When detecting emotions in texts, our results show how EmoAtlas can rival with out-of-box transformer architectures (i.e., transformers pre-tuned for emotion detection on other emotional datasets or LLMs). EmoAtlas is up to 12 times faster than transformers and can support emotions like trust, anticipation and disgust that are rarely supported by transformers. EmoAtlas is not an absolute winner but can win over BERT and BERT-like colleagues in terms of precision and recall on either longer texts, be it Italian Facebook posts or YouTube comments (Sprugnoli, 2020), or shorter texts, like English tweets (Mohammad et al., 2018) and English news headlines (Strapparava & Mihalcea, 2007), as quantitatively measured here. EmoAtlas can even outperform Italian LLMs in detecting surprise, joy and disgust in online posts, although in English LLMs gain roughly 4 points of accuracy above EmoAtlas. This is balanced by EmoAtlas providing interpretable results about the structure of emotional, syntactic and semantic associations between words in texts, which is not possible with black-box (Rudin, 2019) BERT-like models or LLMs. This interplay of valid model performance and lightweight computational cost makes EmoAtlas a valid tool for emotional profiling of textual data across online and offline contexts. Consequently, our key finding is not that a hybrid and interpretable model is better than transformers but rather that such different paradigms can provide comparably good results across varied settings at a mere fraction of the time/computational cost of more elaborate transformer models or LLMs requiring expensive trainings, high-end computational setups or API costs. EmoAtlas might thus be appealing for researchers more interested in lightweight, easy-to-interpret text analysis tools.

Consequently, we show a concrete application of EmoAtlas in the psychometric task of measuring and understanding human creativity ratings in short texts. We find EmoAtlas performs equivalently to the more sophisticated BERT in predicting creativity ratings when using only three network measures and five emotional z-scores. Combining the two models leads to even better performance, suggesting that the semantic divergence measured by BERT might co-exist with a syntactic/emotional divergence of word associations in short stories, as captured by EmoAtlas. Our results thus underline the strong relevance of EmoAtlas for conducting quantitative, yet computationally inexpensive, psychological inquiries of textual data.

Why does EmoAtlas rely on syntactic parsing rather than simpler word co-occurrences? The latter indicate words appearing one after the other or separated by at most m other words, in text. N-gram analysis counts the frequency with which N words appear together in text (e.g., ”good politician”) and co-occurrence networks translate these frequencies into connections between words. These approaches were quite powerful in identifying stances (Mohammad et al., 2016; Stella et al., 2018) and other phenomena like writing style and text authorship (Quispe et al., 2021). However, co-occurrence networks represent only an approximation of syntactic relationships between words in text and their adoption as psychological models of associative knowledge has been met with some criticism (Ninio, 2014; Zweig, 2016). The problem with N-gram analysis is that the combinations of words to be searched in text grow exponentially with N, making it difficult to understand which sequences of words encode specific aspects of stances. Co-occurrence networks can fail at grasping syntactic links between words far apart in sentences (e.g., in ”climate change is a terrible, catastrophic, problematic threat”, ”threat” specifies syntactically ”change” and yet the two words are separated by many others and thus do not co-occur together). These limits pushed computer and cognitive scientists to search for other ways to capture the structure of conceptual associations embedded in texts.

The hybrid synergy of its psychologically validated data and AI, makes EmoAtlas rather unique. The AI is not used to encode vectorial representations of words, as it happens in transformers (Devlin et al., 2018), but rather to detect how concepts are syntactically entwined in texts. This unveils the associative structure of conceptual associations embedded in texts and confers EmoAtlas interpretability for tackling stance detection: Differently from black-box NLP models (Rudin, 2019; Hao et al., 2021), EmoAtlas enables access to the structure of concept associations and semantic frames across thousands of texts in a few seconds (see the computational costs in Fig. 5). In the example about star ratings reported here, this makes it possible for the experimenter to understand that unsatisfied customers found bugs in their beds (see Fig. 6) and this might be one cause for their emotionally neutral profiling of ”bed” in hotel reviews. Similar experiments and additional hypothesis testing might be implemented via emotional analysis and network analysis, see past approaches about COVID-19 (Semeraro et al., 2022; Stella et al., 2020), dialogues in movies (Hipson & Mohammad, 2021; Kumar et al., 2022), suicide letters (Teixeira et al., 2021) or online discourse (Stella et al., 2018; Stella, 2022).

The hybrid nature of EmoAtlas makes it validated from a psychological perspective and independent from training on pre-labeled data. This element stems from the fact that EmoAtlas maps words into emotions according to a psychological mega-study with a large population (Mohammad & Turney, 2013; Mohammad & Kiritchenko, 2015). Emotional profiles are thus entirely dependent on word presence/absence in text and not on opaque, black-box training outcomes, often inaccessible to researchers (Rudin, 2019). The psychological grounding of EmoAtlas confers it a human-centric feature, which is becoming increasingly desirable (Bryson & Theodorou, 2019) for cutting-edge tools in AI. An AI is human-centric if its data modeling and outputs account for the complexity of human psychology and behavior: A model outlining only the presence or absence of an emotion but not its intensities or the semantic frame/context where it appears is not human-centric (Abdul-Mageed & Ungar, 2017; Bryson & Theodorou, 2019; Mohammad &

Kiritchenko, 2018; Mokryn & Ben-Shoshan, 2021; Mohammad & Kiritchenko, 2015). Z-scores and TFMNs in EmoAtlas are deeply entwined with psychological data, enabling reconstructions of conceptual associations that capture the nuance of human perceptions, behaviors, and expressions. Future research might use next-generation lexicons like the Affect Intensity Lexicon (Mohammad & Kiritchenko, 2018) to better quantify the intensity of emotions in texts, even beyond the z-scores already implemented in EmoAtlas. This is the first of a series of future research directions, which we discuss in combination with current limitations.

Limitations and future research

Importantly, both transformers and EmoAtlas could be made better at detecting emotions. BERT (Devlin et al., 2018), its analogs (Sanh et al., 2019; Acheampong et al., 2021) and even LLMs (Wang et al., 2024) can undergo additional training on further datasets and acquire additional insights about emotional features of texts in a specific domain. However, this re-training (Acheampong et al., 2021; Rudin, 2019): (i) is not always possible unless the amount of data to be analyzed is in terms of thousands of texts, (ii) comes at considerable costs for achieving delicate hyperparameter tuning, and (iii) alters the performance of a transformer network over unseen texts compared to reference values from the out-of-box case, potentially conflating overfitting. These aspects might be relatively irrelevant in corporate scenarios, where the focus is achieving outputs as accurately as possible in a restricted range of applications, but they evidently undermine model interpretability in research investigations, i.e., how is the transformer model capturing information from texts of a given domain? With a lexicon approach, the experimenter has fully transparent access to the jargon being captured by EmoAtlas and can also access the syntactic structure embedded in texts via textual forma mentis networks (TFMNs) (Stella, 2020). This synergy makes it easier for the experimenter to test which domain-specific words might be added to the lexicon, following the same protocol outlined in EmoLex (Mohammad & Turney, 2013), and ultimately improving EmoAtlas' performance.

Recent works (Hou & Chen, 2019; Acheampong et al., 2021; Devlin et al., 2018; Basile et al., 2021) have shown that transformer networks can do way better than lexicon-only approaches in the presence of slang or context-dependent words (changing their meaning according to their contexts): BERT and its analogs might capture the meaning of slang words directly from their context of usage whereas EmoAtlas in its bag-of-words mode would fail at reconstructing the emotions specifically evoked by slang (e.g., ”goldfish” is slang for a forgetful person but might have been interpreted as an animal in EmoLex). In these instances, the structured mode of EmoAtlas should be used, instead, as outlined here in the word-level emotional analysis subsection in the Results. EmoAtlas can build a TFMN to infer the emotions attributed to the slang itself by its associates. For instance, in texts reporting ”goldfish” as slang, ”goldfish” itself might be attributed to a semantic frame populated by sad or angry jargon, portraying reactions to forgetfulness and better capturing the meaning of the slang itself. The ability of EmoAtlas to perform emotional analysis based on AI-driven syntactic relationships makes it suitable for contextual emotional shifting (Turney & Mohammad, 2019; Abramski et al., 2023; Li et al., 2020). In the presence of fine-grained, topic-matched, future corpora with word-level emotional profiles, future research might systematically compare EmoAtlas and transformer networks in their ability to understand words' emotions according to their context. Another relevant direction for future developments would be adding entity recognition to the current abilities of EmoAtlas, e.g., to identify how entities expressed by more than one word, like ”bubble tea” or ”New York City”, could be framed within texts. This would require the adoption of more advanced techniques, potentially leaning on large language models (Hassani et al., 2020; Wang et al., 2024), and thus represent another opportunity for synergetic yet interpretable combinations of LLMs and EmoAtlas.

Conclusion

EmoAtlas is a hybrid tool for emotional analysis via psychological emotional data, automatic syntactic parsing and interpretable network science construction. Our tests across different datasets of short and long texts report that EmoAtlas performs as well, if not better, than out-of-box (i.e., pre-tuned) transformer networks. In producing emotional profiles, EmoAtlas can be 12 times faster than BERT and its analogs. Giving direct access to the network structure of conceptual associations, EmoAtlas provides experimenters with a content mapping tool that can investigate how concepts were framed in thousands of texts, within mere seconds. Currently supporting meaning negations, idiomatic expressions and word-level context-dependent emotional analysis, now running also on accessible web services like Google Colab, we believe EmoAtlas represents a suitable tool for the emotional investigation of text corpora, be it in online social discourse, corporate settings or information management channels.

Methods

This section outlines the key datasets and methodologies implemented in EmoAtlas. The library is available for download online to ensure reproducibility of results.

Text analysis in EmoAtlas

Figure 1 reports an example of the data analysis pipelines implemented in EmoAtlas: The tool can output data visualizations reporting emotional profiles, with statistical testing (top left), and textual forma mentis networks (top right), visualized in a convenient edge-bundling layout. Let us outline in more detail how the pipelines were implemented. EmoAtlas reads a text in input and performs the following steps:

1. Sentence splitting: Each text is separated into sentences through punctuation analysis in spaCy (Honnibal & Montani, 2017). Sentences are then analyzed independently from others.
2. Text tokenization: Each sentence is transformed into a sequence of tokens, i.e., words, punctuation and special characters (e.g., emojis). Despite punctuation being relevant in some cases, e.g., writing style analysis (Darmon et al., 2021), we discard it here and focus only on words and other tokens with emotional content like emojis.
3. Text normalization: Word tokens are normalized. The user can decide to normalize text via either stemming (e.g., happier → happi, happiness → happi) or lemmatization (e.g., happier → happy, happiness → happiness). Lemmatization is performed through spaCy (Honnibal & Montani, 2017). Stemming is performed through Snowball in nltk as in other approaches (Colladon et al., 2020). At this stage, emojis are translated into English words according to Emojipedia (see $https://emojipedia.org/about/$, last access: 19/10/2022).
4. Lexical analysis in ”bag-of-words mode”: EmoAtlas is a hybrid lexicon-based approach, working either as a bag-of-words model or by giving structure to text. In the ”bag-of-words mode”, lexical analysis counts how many words and emojis in each sentence elicit a certain emotion according to EmoLex (Mohammad & Turney, 2013). These word counts are then summed up across all sentences in a text and used to perform emotional analysis (see Methods) and Fig. 1 (top left).
5. Support for idiomatic n-grams (both in ”structured” and ”bag-of-words” modes): Some idiomatic tokens might need to be considered together to convey meaning, e.g., the bigram ”Santa Claus” can provide emotions of joy and trust that are absent in its constituting tokens ”santa” and ”claus”. EmoAtlas supports a dictionary-based set of idiomatic expressions for all the 18 languages currently supported (see Introduction).
6. Lexical analysis in structured mode: This mode gives structure to text through syntactic parsing (Ferrer-i Cancho et al., 2022), as implemented in spaCy (Honnibal & Montani, 2017). This task is carried over by an AI pre-trained on a vast amount of linguistic data to spot syntactic relationships between words in sentences, an example is reported in Fig. 1 (bottom). By default, EmoAtlas uses the $en_core_web_lg$ language model (Honnibal & Montani, 2017), a recurrent neural network architecture trained to spot syntactic relationships in over 1.4 million annotated transcripts and achieving an accuracy around 90%, cf. Hovy et al. (2006). Each sentence is transformed into a syntactic tree, where words are automatically tagged according to their part-of-speech features (e.g., nouns are labeled as NOUN). The user can define stopwords according to specific labels. By default, non-stopwords are considered being nouns, verbs, adverbs and pronouns, some psychology studies might be interested in self-perceptions, so ”I”, ”me”, ”myself”, etc. would be words worth of investigation (Teixeira et al., 2021). Auxiliary verbs are also checked for negations of meaning. On the syntactic tree (see Fig. 1), the distances t_{ij} separating non-stopwords i and j are measured. Every sampled i and j are linked if $t_{ij} < K$, giving rise to a textual forma mentis network (TFMN) (Stella, 2020; Semeraro et al., 2022; Stella, 2022). By default K is set to 3, thus measuring local syntactic relationships on the syntactic tree (Ferrer-i Cancho et al., 2022). A TFMN is built for each sentence. Iterating through sentences, TFMNs are merged by joining their edge lists and considering a simple graph (Newman, 2018).
7. TFMN visualization (only in structured mode): In the TFMN (see Fig. 1), stopwords do not appear and other concepts are linked despite stopwords not being included. Cyan (red) indicates positive (negative) words as outlined in EmoLex (Turney & Mohammad, 2019). Cyan (red) links connect positive words. Purple (gray) links connect positive and negative (neutral) words. Textual forma mentis networks represent proxies of the syntactic and semantic relationships between words. Semantic relationships are added on top of syntactic links, i.e., semantic relationships can link only pairs of words in the same syntactic network. Semantic relationships express overlap of meaning, i.e., synonyms, according to WordNet (Miller, 1995). The user can highlight them in green. TFMNs make it possible to identify which concepts ”keep company” to target ones, constituting the semantic frame of word associations surrounding a target word (Fillmore & Baker, 2001). Importantly, semantic frames can be emotionally profiled, see also the emotions of the semantic frame of ”vaccine” in Semeraro et al. (2022) or of ”gender” in Stella (2020). Negated words are accounted for in emotional analysis for semantic frames/network neighborhoods. Such profiling works as in the ”bag-of-words” mode but considers only words selected through network structure.
8. Negation of meaning (only in structured mode): Syntactic tree parsing enables a better control of meaning negations in texts compared to word co-occurrence anal-

ysis, e.g., see the implementation of VADER (Hutto & Gilbert, 2014). For instance, ”this is not very cool” negates the meaning of ”cool”. However, the negation ”not” and ”cool” itself are not adjacent tokens in the sentence and could thus be ignored by word co-occurrence analysis despite being syntactically linked in the sentence. In structured mode, we test whether any sentence contains a negator, i.e., ”not” (”not” or ”cannot”) or ”n't” (e.g., ”don't”). On the tree, we check the auxiliary verb the negation depends syntactically on (i.e., ”is” in ”this is not very cool”). Always on tree, we check for the nearest neighbors depending on the auxiliary verb through complement specification links, see the categorization in spacy (Honnibal & Montani, 2017). We negate each of these complements (e.g., ”cool”). It can occur that complements are linked through contrastive prepositions – like ”but” – which also negate meaning, e.g., ”this is not very cool but still right”. Through syntactic parsing we can check for the occurrence of contrastive prepositions and avoid negating their depending complements, i.e., negating ”cool” but not ”right”. Negation takes place by substituting a word for its antonym as encoded in WordNet (Miller, 1995). Antonyms are selected from a 1-to-1 table, neglecting context. Future implementations of EmoAtlas might use BERT for context-dependent antonym selection (Acheampong et al., 2021) but we decided not to use it for the sake of a fair comparison between EmoAtlas and transformers. Words that should be negated but without an antonym in the list are transformed into new tokens, e.g., ”not_discombobulated”, so that they do not enter in emotional analysis for semantic frames. Original words are replaced by their negations. Network visualizations report only negated words, providing a more faithful representation of negated and assertive associative knowledge expressed in texts.

Multi-language support in EmoAtlas

EmoAtlas supports emotional analysis via syntactic parsing and lemmatization in 18 different languages: Catalan, Chinese, Danish, Dutch, English, French, German, Greek, Italian, Japanese, Lithuanian, Macedonian, Norwegian, Polish, Portuguese, Romanian, Russian and Spanish. The user can decide to normalize text via either lemmatization (e.g., happier → happy, happiness → happiness) or stemming (e.g., happier → happi, happiness → happi). EmoAtlas provides stemming support for 12 languages: Danish, Dutch, English, French, German, Italian, Norwegian, Portuguese, Romanian, Russian and Spanish.

In the current manuscript, we test EmoAtlas in English and in Italian given the language proficiency of the authorial team. We encourage authors proficient in other languages to test the emotional lexicons and EmoAtlas network-building abilities through future research.

Emotion detection in EmoAtlas: Emotional profiles

As in previous works, we define an emotional profile as a set of emotions attributed to a given text or word. In EmoAtlas, text-level emotional profiling is defined in terms of considering how many words eliciting a given emotion are contained in a text. Word-level emotional profiling of texts is defined in terms of considering how many emotion-eliciting words are linked to the target word through syntactic or semantic links in text. Notice that word-level emotional profiling accounts also for meaning negation.

The emotions composing a given profile are the same ones reported in EmoLex and include basic building-block of several more complex emotional states according to Plutchik's theory of emotional processing to threats (Plutchik, 2003):

- **Anger**, a negative emotion representing reactions of irritation and rage towards an external threat;
- **Disgust**, a negative emotion indicating aversion and closure;
- **Fear**, a negative emotion indicating a need to actively avoid and prevent potential threats;
- **Trust**, a positive emotion indicating openness towards the outer world;
- **Joy**, a positive emotion of excitement and satisfaction;
- **Sadness**, a negative emotion indicating states of sorrow, thoughtfulness and inhibition;
- **Surprise**, a neutral emotion relative to being upset or startled by an unexpected event;
- **Anticipation**, a neutral emotion indicating one's projection into future events, including desire or anxiety.

The above emotions can be combined to obtain more complex states, e.g., anticipation and trust together make hope. For a more detailed description of emotions, please see Ekman and Davidson (1994) and Mohammad and Turney (2013). The mega-study by Mohammad and colleagues is also known as the Word-Emotion Association Lexicon, indicating how human participants associated individual concepts with emotional states. In other words, the resulting dataset indicated which emotions were elicited or *evoked* in human subjects reading individual concepts. The study used crowd-sourcing on Mechanical Turk for achieving a large-scale mapping of English words (including 14,000 unique lemmas) and was also used in other successful investigations about affect patterns, cf. Mohammad and Turney (2013).

Emotion detection in EmoAtlas: Null models

Comparing the emotional richness of different texts is problematic because they might involve different numbers of words. Comparing the observed emotional profile of a text with a null model represents a more appropriate comparison, especially considering that there are more words eliciting some emotions (e.g., fear and joy) rather than others (e.g., anticipation and sadness) in the EmoLex (Mohammad & Turney, 2013). On the one hand, using only out-of-box scores can answer questions like "Did a portion of text use words associated with a certain emotion?". On the other hand, a statistical comparison can answer questions like "Was trust the most prominent emotion just because there are more words eliciting trust in the reference texts? Does the target text include a lot more joyful words than the reference texts? etc.". To implement statistical testing, we consider z-scores:

$$z_e(T) = \frac{n_e(T) - \langle r_e(T) \rangle}{\sigma_{r_e(T)}}, \tag{1}$$

Relative to the count $n_e(T)$ of emotional words eliciting emotion e in text T, the expected number $\langle r_e(T) \rangle$ of words eliciting the same emotion when drawing at random a sample of M words, and the standard deviation $\sigma_{r_e(T)}$ for $r_e(T)$ counts over N samples. M is fixed being the number of words eliciting any (i.e., at least one) emotion in T. N can be tuned to reduce fluctuations over the estimates of $z_e(T)$. The default value is set to $N = 300$. Words can be sampled uniformly at random from either EmoLex or another reference corpus. The default behavior of EmoAtlas is to draw words uniformly at random from EmoLex, so that a random sample can reflect the unequal numbers of words eliciting different emotions (e.g., the dataset features more words inspiring trust rather than surprise). This behavior accounts for the larger variety of words expressing negative (compared to positive) emotions in language, as reflected by the EmoLex (Mohammad & Turney, 2013). However, the user of EmoAtlas can also change this behavior and consider custom reference corpora, like general-purpose corpora of word frequency, which might account for the positivity bias in English, i.e., words with positive valence occurring more frequently in large corpora (Dodds et al., 2015). Custom-based repositories of news media articles were used as reference models in EmoAtlas in a previous work (Semeraro et al., 2022). Z-scores and custom reference baselines might also be used to compare directly any two texts (Semeraro et al., 2022).

Exceptions to statistical testing

When m is relatively low, distortions over the distributions of words eliciting emotions in a given text make it difficult to apply statistical testing based on z-scores (Stella, 2022). EmoAtlas considers specific exceptions where z-scoring is not adopted. When $m = 0$, the text is reported as being emotionally neutral and its z-score emotional profile is an 8-D vector of zeros. When $m = 1$, the z-score emotional profile in output has a z-score fixed to 2 only for those emotions elicited by the single word evoking emotional states (e.g., "die!" elicits anger). When $2 \leq m \leq 5$, the z-score emotional profile in output has a z-score fixed to 2 only for those emotions elicited by at least two words evoking emotional states (e.g., "live or die" does not elicit anger, whereas "fight and die!" does). For other cases, emotional profiling with statistical sampling is used and users are invited to check carefully their z-scores for sample size biases.

Emotional flowers

EmoAtlas produces visualizations called emotional flowers for their resemblance with flower petals (Plutchik, 2003; Semeraro et al., 2021; Stella, 2020). In this visualization (see Fig. 1 top), the rejection region relative to $|z| < 1.96$ is a grey disk. Bars are shaped like petals, one per emotion, and colored differently. Those petals going out of the rejection region indicate a semantic frame with jargon eliciting a given emotion more than random expectation (within a significance level of 0.05). Petals contained below the rejection region indicate that the analyzed text was poorer than expected in words eliciting those emotions. The significance level can be tuned by the experimenter.

Speed boosting emotional profiling with look-up tables

The statistical testing of emotional profiles requires repeated sampling words at random, either from EmoLex or from a baseline corpus, like a collection of newspapers, cf. Semeraro et al. (2022). When the number of words eliciting at least one emotion, m, is fixed, the analytical computation of the expected average emotional random scores $\langle r_e(T) \rangle$ is complicated by the unknown of how many emotions a single word can elicit, e.g., "war" can elicit disgust, fear, and anger at the same time. This makes computational sampling quite relevant for statistical testing.

To speed up statistical testing, we endowed EmoAtlas with adaptive look-up tables, with pre-compiled expected scores $\langle r_e(T) \rangle$ and standard deviations $\sigma_{n_e(T)}$ for values of m ranging from 1 to 2000 in steps of 1. Once EmoAtlas identifies the number of words eliciting at least one emotion – m itself – rather than performing 300 random samplings, the pre-compiled values are retrieved from the look-up table, instead. This takes place only when sampling from EmoLex is selected by the user (this is a default setting in the current implementation).

The look-up table is adaptive in the sense that if m from a given text falls outside of the above range, sampling from EmoLex is performed and a new entry, for that value of m, is added to the table, including expected scores $\langle r_e(T) \rangle$ and standard deviations $\sigma_{n_e(T)}$. An adaptive look-up table is created from scratch for every other baseline selected by the user. The more texts get compared against such baseline (e.g., another corpus of news articles), the more entries with different values of m will be created and the faster will EmoAtlas get in producing emotional profiles over that baseline.

Emotion detection in EmoAtlas: Reference datasets in English and Italian

We test EmoAtlas in emotional profiling against out-of-box versions of BERT, RoBERTa, AlBERT, and DistilBERT in either English or Italian text corpora. These out-of-box models underwent some fine tuning for emotion detection but on other emotional datasets compared to the ones tested here.

In English we used the following datasets:

1. GoEmotions (Demszky et al., 2020), a manually annotated corpus of almost 58,009 Reddit posts, each one labeled for one or more 27 emotional states or the ”neutral” tag. For the comparison, we focused on emotions overlapping with EmoLex, i.e., joy, anger, sadness, surprise, disgust, and fear. We discarded all other emotional states since they did not map clearly into basic emotional states. This selection led to a dataset with 58,011 posts. Each post was rated by at least three and at most five individual raters. Posts were relatively short, containing on average 22.69 words.
2. Annotated tweets from SemEval-2018 Task 1 – Emotion Classification task (Mohammad et al., 2018), a corpus of 6838 tweets (training set only). Each tweet was manually annotated to describe the mental state of the author according to 11 emotional states, i.e., eight from EmoLex and also pessimism, optimism and love.
3. 250 annotated news titles, extracted from either online websites (e.g., CNN) or newspapers as introduced in the SemEval-2007 Task 14. These headlines were rated continuously on a range between 0 and 100, according to the emotional intensity elicited by a given headline (0 means no emotion, 100 indicates a strongly present emotion). Six annotators independently labeled each headline for six different emotions, i.e., all the ones contained in EmoLex except trust and anticipation.

In Italian we used the following datasets:

1. FEEL-IT (Bianchi et al., 2021), a corpus containing 2037 Italian tweets annotated by human coders in terms of sadness, anger, fear, and joy. Each tweet contains on average 10 ± 2 words.
2. MultiEmotions-it (Sprugnoli, 2020), a manually annotated corpus with 3,240 comments in Italian relative to YouTube videos and Facebook advertisements. Each comment was rated by two individuals with labels expressing the eight categorical emotions in EmoLex (Mohammad & Turney, 2013) plus ”sarcasm”. Each comment contains on average 20 ± 2 words.

Emotion detection: Using emotional scores and *z*-scores

To classify the presence or absence of a given emotion in a text we use two methods: (i) checking the emotional scores produced out-of-box from BERT, BERT-like models, or (ii) checking the z-scores relative to a given text when compared against a given baseline.

All BERT-like models produce a confidence value, indicating how confident is the transformer network that the outputted emotional label of the text was ”correct” (Demszky et al., 2020). This continuous value, constrained between 0 and 1, can be estimated by transformer networks thanks to their training on pre-labeled data (Acheampong et al., 2021; Sprugnoli, 2020). For BERT and BERT-like models, we consider this value to be an emotional score. Given its probabilistic interpretation as a ”confidence” value, c_e, only if $c_e(T) > 0.5$ then we consider the text T as eliciting that given emotion e. Else, if $c_e(T) \leq 0.5$, we consider T as not eliciting the emotion e. The threshold 0.5 rises naturally as the median of the [0,1] range where confidence is contained by design, within transformer networks (Acheampong et al., 2021). In EmoAtlas, emotional scores might be considered as the raw word counts of emotional words in a given text. These scores would, however, depend on the length of text and the number of emotions present in it, considerably complicating the normalization of these scores compared to transformers. In the absence of a clear way for normalizing such scores, we resorted to considering only z-scores in EmoAtlas, as to reduce potential normalization biases (e.g., a lack of information about how to translate the information on the 0.5 threshold from transformers into EmoAtlas).

Z-scores can be computed by EmoAtlas as described in the Methods subsection ”Emotion detection in EmoAtlas: Null models”. As a baseline we used the default baseline of EmoAtlas, i.e., drawing words at random from EmoLex. Z-scores can be computed also for BERT-like models. Consider a corpus $C = \{T_i\}_i$ of texts. Call a BERT-like model on all the elements of C and then partition it into two balanced partitions, C_1 and C_2. For attributing z-scores to texts in C_1, we considered C_2 as a baseline and computed all emotional scores/confidence values for the presence of a given emotion e. We thus obtained a set of confidence values that we used as a baseline, computing its average and standard deviation. For each text of C_1, we compared its confidence value to the average and standard deviations obtained from C_2, ultimately getting a z-score. We repeated this operation by switching C_1 and C_2 (i.e., subsequently using C_1 as a baseline for attributing z-scores to texts in C_1).

Prompts for emotion detection with LLMS

We used LLMs like ChatGPT 3.5 in its turbo-0125 version (via API, in English and Italian), LLaMAntino 2-Chat-13b-HF-Q8 (in Italian) and Mistral 7B-instruct-v0.2-Q8 (in English). To instruct LLMs to perform emotion detection, we wrote the following 2 prompts, one for English and one for Italian.

- The English one mentions: ”Below is an instruction that describes a task. Write only a Python list that appropriately completes the request. You cannot write anything other than a Python list. Consider the task of detecting emotions in texts. Consider Plutchik’s theory according to which text can elicit ONLY THESE 8 emotions: joy, anger, sadness, anticipation, fear, trust, surprise, and disgust. Which of these eight emotions are encoded in the TEXT BELOW? [TEXT HERE] In writing your response you must not use any word outside of: ”joy”, ”anger”, ”sadness”, ”anticipation”, ”fear”, ”trust”, ”surprise”, ”disgust”.
- The Italian prompt mentions: ”Di seguito viene riportata un’istruzione che descrive un compito. Scrivi solo una lista Python che completi in modo appropriato la richiesta. Non puoi scrivere nulla di diverso da un elenco Python. Considera il compito di rilevare le emozioni nei testi. Considera la teoria di Plutchik secondo cui un testo potrebbe suscitare SOLO QUESTE 8 emozioni: joy, anger, sadness, anticipation, fear, trust, surprise, disgust. Quali di queste 8 emozioni sono codificate in QUESTO TESTO: [TEXT HERE] Nello scrivere la tua risposta non devi usare nessuna parola al di fuori di: ”joy”, ”anger”, ”sadness”, ”anticipation”, ”fear”, ”trust”, ”surprise”, ”disgust”.

We then considered the lists of emotions for each text, discarding those that did not adhere with the required format.

Emotion detection: Model assessment

We test each language model, e.g., EmoAtlas and (Tables 11 and 12) BERT-related transformers, through a classification task where any of the eight emotions covered by EmoAtlas can be present or absent in a given text, from a given corpus. The presence of emotions is coded, as a golden rule, by human raters, as reported in the original papers introducing the datasets (see Methods Section 4.5).

Detecting the presence or absence of an emotion in a text thus becomes a binary classification task. Model performance is measured through multiple metrics like accuracy, precision and recall (Alpaydin, 2020). In a binary classification task, accuracy is the proportion of correct predictions made by the model. Precision is the proportion of positive predictions that are correct. Recall is the proportion of actual positive cases that are correctly predicted by the model. We do not report an F1 score (the harmonic mean of precision and recall) in the tables because of the considerable variance between the two in our tasks.

Artificial psychometrics: EmoAtlas and creativity ratings

As reported in the Results, we tested linear regression, multilayer perceptrons and random forests, as implemented in Mathematica 11.3, but found that random forests achieved the highest correlations between predictions and empirical test values, within a fourfold cross-validation of mean creativity ratings.

Among the tested features, we focused on structural ways of assessing concept connectivity, guided by Mednick’s hypothesis of creativity as a psychological construct rising from the ability to associate concepts being more remotely connected within the mental associative knowledge available to an individual (Mednick, 1962). Over textual forma mentis network representations of short stories, built through EmoAtlas, we measured:

- Average graph distance, i.e., the average smallest number of links separating any two nodes (Newman, 2018). We focused only on the largest connected components of each network, i.e., the largest set of nodes connected by at least one sequence of links. A larger average graph distance indicates a way to associate concepts through more links, i.e., through longer/more remote associations.
- Mean local clustering coefficient, i.e., the average fraction of neighbors of nodes which share a link with each other (Newman, 2018). A higher mean local clustering coefficient means that concepts tend to be more cohesively linked/clustered with each other.
- Inverse graph diameter, i.e., the inverse of the largest distance between any two nodes in a network (Newman, 2018). In the presence of disconnected nodes, the diameter is assumed being infinite (Newman, 2018). For connected networks, larger values of the diameter indicate an ability to link concepts via more remote associations.

Library availability, code, and gentle online tutorials

EmoAtlas can be downloaded online from GitHub: GitHub Link (accessed: 30/05/2024). A Google Colab version of EmoAtlas, enabling text analysis without the need for local installations or package download, is available here: Google Colab Link (accessed: 30/05/2024). Online tutorials, in the format of Jupyter Notebooks, are available here: Gentle Starting Guide and Gentle Advanced Guide (for network analysis).

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Availability of Data and Materials All code and data are available online, see the Section "Library availability, code, and gentle online tutorials".

Code Availability All code and data are available online, see the Section "Library availability, code, and gentle online tutorials".

Declarations

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